

# Rapport-preuve - KGE Models (provenance : évalué vs mentionné)

Corpus : **55 articles**. Évalués validés vs tables\_only, mentionnés vs no\_tables (prose). Matching déterministe.

## Scores

| Score                                  | Valeur       |
|----------------------------------------|--------------|
| Précision évalués                      | <b>94.5%</b> |
| Précision mentionnés                   | <b>97.9%</b> |
| Recall évalués (indicatif)             | <b>89.5%</b> |
| Recall mentionnés (indicatif)          | <b>95.5%</b> |
| Évalués vérifiés / mentionnés vérifiés | 344 / 238    |

## A. Preuve - modèles/méthodes ÉVALUÉS (344)

| Article         | Entité       | Trouvé | Extrait (tableau)                                |  |  |
|-----------------|--------------|--------|--------------------------------------------------|--|--|
| AdaptativeNS.md | ComplEx      | oui    | $\text{Re}(\mathbf{h}^T \mathbf{t})$             |  |  |
| AdaptativeNS.md | ConvE        | oui    | $\vec{f}(\mathbf{r},$                            |  |  |
| AdaptativeNS.md | DistMult     | oui    | $\mathbf{h}^T \mathbf{t}$                        |  |  |
| AdaptativeNS.md | MuRP         | oui    | $\mathbf{d}_{\mathbf{B}}(\exp_{\mathbf{B}}$      |  |  |
| AdaptativeNS.md | RotatE       | oui    | $\mathbf{h} \odot \mathbf{m}$                    |  |  |
| AdaptativeNS.md | TransE       | oui    | $d_{0.226}(\mathbf{h}, \mathbf{t})$              |  |  |
| ADNS.md         | DistMult     | oui    | e 3. Evaluation of Negative Sampling on DistMult |  |  |
| ADNS.md         | TransE       | oui    | 2.** Evaluation of Negative Sampling on TransE   |  |  |
| ASA.md          | ComplEx      | oui    | 53.18(10)                                        |  |  |
| ASA.md          | ConvE        | oui    | 49.65(11)                                        |  |  |
| ASA.md          | DistMult     | oui    | 53.94(9)                                         |  |  |
| ASA.md          | GATNE        | oui    | 96.25(5)                                         |  |  |
| ASA.md          | HAN          | oui    | 87.57(7)                                         |  |  |
| ASA.md          | metapath2vec | oui    | 94.15(6)                                         |  |  |
| ASA.md          | R-GCN        | oui    | 97.16(4)                                         |  |  |
| ASA.md          | RelGNN       | oui    | 98.84(2)                                         |  |  |

| Article    | Entité                   | Trouv é | Extrait (tableau)                                                                    |  |  |
|------------|--------------------------|---------|--------------------------------------------------------------------------------------|--|--|
| BatchNS.md | ComplEx                  | oui     | gt;</td> </tr> <tr> <td>ComplEx</td><br><td>x ■ θ■</td> <t                           |  |  |
| BatchNS.md | DeepWalk                 | oui     | </thead> <tbody> <tr><br><td>DeepWalk*</td> <td>0.691</td><br><t                     |  |  |
| BatchNS.md | HolE                     | oui     | 749</td> </tr> <tr> <td>HolE (Nickel<br>et al., 2016b)</td> <td>                     |  |  |
| BatchNS.md | MILE (1 level)           | oui     | GB</td> </tr> <tr> <td>MILE (1<br>level)*</td> <td>0.629</td> <                      |  |  |
| BatchNS.md | MILE (5 levels)          | oui     | GB</td> </tr> <tr> <td>MILE (5<br>levels)*</td> <td>0.505</td> <                     |  |  |
| BatchNS.md | PBG                      | oui     | GB</td> </tr> <tr> <td>PBG (1<br>partition)</td> <td>0.749</t                        |  |  |
| BatchNS.md | PBG (ComplEx)            | oui     | 785</td> </tr> <tr> <td>PBG<br>(ComplEx)</td> <td>0.242</td> <t                      |  |  |
| BatchNS.md | PBG (TransE)             | oui     | 910</td> </tr> <tr> <td>PBG<br>(TransE)</td> <td>0.265</td> <t                       |  |  |
| BatchNS.md | R-GCN+                   | oui     | 840</td> </tr> <tr> <td>R-GCN+<br>(Schlichtkrull et al., 2018)</td>                  |  |  |
| BatchNS.md | Reciprocal<br>ComplEx-N3 | oui     | 838</td> </tr> <tr> <td>Reciprocal<br>ComplEx-N3 (Lacroix et al.,<br>2018)</td> <td> |  |  |
| BatchNS.md | RESCAL                   | oui     | </thead> <tbody> <tr><br><td>RESCAL</td> <td>A■x</td><br><td>&                       |  |  |
| BatchNS.md | StarSpace                | oui     | 842</td> </tr> <tr> <td>StarSpace<br>(Wu et al., 2018)</td> <td>.</t                 |  |  |
| BatchNS.md | TransE                   | oui     | gt;</td> </tr> <tr> <td>TransE</td><br><td>x + θ■</td> <t                            |  |  |
| CAKE.md    | HAKE                     | oui     | ng></td> </tr> <tr> <td>HAKE</td><br><td>34</td> <td>0.                              |  |  |
| CAKE.md    | RotatE                   | oui     | ng></td> </tr> <tr> <td>RotatE</td><br><td>35</td> <td>0.                            |  |  |
| CAKE.md    | TransE                   | oui     | </thead> <tbody> <tr><br><td>TransE</td> <td>35</td> <td>0.                          |  |  |
| CANS.md    | CKRL                     | oui     | 9.9</td> </tr> <tr><br><td>CKRL(LT)</td> <td>230</td> <                              |  |  |
| CANS.md    | TransE                   | oui     | </thead> <tbody> <tr><br><td>TransE</td> <td>249</td> <td>1                          |  |  |
| CCS.md     | Op-TransD                | oui     | 873</td> </tr> <tr><br><td>Op-TransD</td> <td>3352</td><br><td>                      |  |  |
| CCS.md     | Op-TransE                | oui     | </thead> <tbody> <tr><br><td>Op-TransE</td> <td>4171</td><br><td>                    |  |  |
| CCS.md     | Op-TransH                | oui     | 852</td> </tr> <tr><br><td>Op-TransH</td> <td>4404</td><br><td>                      |  |  |
| CCS.md     | Op-TransR                | oui     | 815</td> </tr> <tr><br><td>Op-TransR</td> <td>3685</td><br><td>                      |  |  |
| CCS.md     | TransD                   | oui     | g functions for TransE, TransH,<br>TransR, TransD. <table> <thead><br><tr> <th       |  |  |

| Article            | Entité   | Trouv é | Extrait (tableau)                                                                        |  |  |
|--------------------|----------|---------|------------------------------------------------------------------------------------------|--|--|
| CCS.md             | TransE   | oui     | 4. Definition of scoring functions for TransE, TransH, TransR, TransD. <table> <thead>   |  |  |
| CCS.md             | TransH   | oui     | inition of scoring functions for TransE, TransH, TransR, TransD. <table> <thead>         |  |  |
| CCS.md             | TransR   | oui     | f scoring functions for TransE, TransH, TransR, TransD. <table> <thead> <tr>             |  |  |
| ConceptDriven.md   | DistMult | oui     | >81.4</td> <td>4.15</td> </tr> <tr> <td>DistMult [18]</td> <td>88.4</td> <td>86.1</td> < |  |  |
| ConceptDriven.md   | DKRL     | oui     | d>44.5</td> <td>52.2</td> </tr><tr> <td>DKRL [16]</td> <td>26.0</td> <td>100.00</td>     |  |  |
| ConceptDriven.md   | RotatE   | oui     | d>43.2</td> <td>50.7</td> </tr><tr> <td>RotatE [17]</td> <td>47.6</td> <td>100.00</td>   |  |  |
| ConceptDriven.md   | TransE   | oui     | 0↑</th> </tr> </thead> <tbody> <tr> <td>TransE [3]</td> <td>24.3</td> <td>100.00</td>    |  |  |
| ConceptDriven.md   | TransH   | oui     | d>40.6</td> <td>47.6</td> </tr><tr> <td>TransH [14]</td> <td>23.1</td> <td>100.00</td>   |  |  |
| ConceptDriven.md   | TransR   | oui     | d>44.5</td> <td>52.2</td> </tr><tr> <td>TransR [15]</td> <td>23.2</td> <td>100.00</td>   |  |  |
| CondConstraints.md | TransE   | NON     | -                                                                                        |  |  |
| dans.md            | ComplEx  | oui     | RGCN) and decoders (DistMult, RotatE or ComplEx). The best results are in bold, and the  |  |  |
| dans.md            | DistMult | oui     | the same backbone (RGCN) and decoders (DistMult, RotatE or ComplEx). The best results a  |  |  |
| dans.md            | RotatE   | oui     | backbone (RGCN) and decoders (DistMult, RotatE or ComplEx). The best results are in bo   |  |  |
| DeMix.md           | ComplEx  | oui     | ng></td> </tr> <tr> <td>ComplEx+Uniform</td> <td>0.429</td>                              |  |  |
| DeMix.md           | DistMult | oui     | </thead> <tbody> <tr> <td>DistMult+Uniform</td> <td>0.412</td>                           |  |  |
| DeMix.md           | HAKE     | oui     | ng></td> </tr> <tr> <td>HAKE+Unifor m<sup> $\Delta$ </td> <td>0.                         |  |  |
| DeMix.md           | RotatE   | oui     | </thead> <tbody> <tr> <td>RotatE+U niform<sup> $\Delta$ </td> <td>0.                     |  |  |
| DeMix.md           | TransE   | oui     | les</td> </tr> <tr> <td>TransE+Uniform</td> <td>0.175*</td>                              |  |  |
| DHNS.md            | AdaMF    | oui     | .88</td> </tr> <tr> <td>AdaMF</td> <td><u>32.51</u></td>                                 |  |  |
| DHNS.md            | ComplEx  | oui     | .80</td> </tr> <tr> <td>ComplEx</td> <td>27.48</td> <td>                                 |  |  |
| DHNS.md            | DistMult | oui     | .68</td> </tr> <tr> <td>DistMult</td> <td>23.03</td> <td>                                |  |  |

| Article     | Entité   | Trouvé | Extrait (tableau)                                                                            |  |  |
|-------------|----------|--------|----------------------------------------------------------------------------------------------|--|--|
| DHNS.md     | GC-OTE   | oui    | .71</td> </tr> <tr> <td>GC-OTE</td><br><td>31.85</td> <td>                                   |  |  |
| DHNS.md     | IKRL     | oui    | wspace="8">MMKGC Models</td><br><td>IKRL</td> <td>26.82</td> <td>                            |  |  |
| DHNS.md     | MMKRL    | oui    | .72</td> </tr> <tr> <td>MMKRL</td><br><td>26.81</td> <td>                                    |  |  |
| DHNS.md     | OTKGE    | oui    | .62</td> </tr> <tr> <td>OTKGE</td><br><td>23.86</td> <td>                                    |  |  |
| DHNS.md     | PairRE   | oui    | .73</td> </tr> <tr> <td>PairRE</td><br><td>31.13</td> <td>                                   |  |  |
| DHNS.md     | RotatE   | oui    | r framework DHNS</mark> integrated<br>with RotatE model and some<br>state-of-the-art baselin |  |  |
| DHNS.md     | RSME     | oui    | .44</td> </tr> <tr> <td>RSME</td><br><td>29.76</td> <td>                                     |  |  |
| DHNS.md     | TBKGC    | oui    | .07</td> </tr> <tr> <td>TBKGC</td><br><td>28.08</td> <td>                                    |  |  |
| DHNS.md     | TransAE  | oui    | .58</td> </tr> <tr> <td>TransAE</td><br><td>28.09</td> <td>                                  |  |  |
| DHNS.md     | TransD   | oui    | .23</td> </tr> <tr> <td>TransD</td><br><td>21.52</td> <td>                                   |  |  |
| DHNS.md     | TransE   | oui    | 7">Unimodal KGC Models</td><br><td>TransE</td> <td>24.86</td> <td>                           |  |  |
| DHNS.md     | VBKGC    | oui    | .83</td> </tr> <tr> <td>VBKGC</td><br><td>30.61</td> <td>                                    |  |  |
| DMNS.md     | ARVGA    | oui    | 005</td> </tr> <tr> <td>ARVGA</td><br><td>.732 ± .011</td>                                   |  |  |
| DMNS.md     | GAT      | oui    | </thead> <tbody> <tr> <td>GAT</td><br><td>.766 ± .006</td>                                   |  |  |
| DMNS.md     | GCN      | oui    | link prediction against baselines using<br>GCN as the base encoder. <table><br><thead>       |  |  |
| DMNS.md     | GraphGAN | oui    | 004</td> </tr> <tr><br><td>GraphGAN</td> <td>.739 ±<br>.003</td>                             |  |  |
| DMNS.md     | GVAE     | oui    | 003</td> </tr> <tr> <td>GVAE</td><br><td><u>.783 ± .003</u></td>                             |  |  |
| DMNS.md     | KBGAN    | oui    | 009</td> </tr> <tr> <td>KBGAN</td><br><td>.615 ± .004</td>                                   |  |  |
| DMNS.md     | SAGE     | oui    | ng></td> </tr> <tr> <td>SAGE</td><br><td>.598 ± .014</td>                                    |  |  |
| DMNS.md     | ScaLed   | oui    | 001</td> </tr> <tr> <td>ScaLed</td><br><td>.676 ± .004</td>                                  |  |  |
| DMNS.md     | SEAL     | oui    | 002</td> </tr> <tr> <td>SEAL</td><br><td>.751 ± .007</td>                                    |  |  |
| DNS.md      | RESCAL   | oui    | parameters. RNS results for TransE<br>and RESCAL are borrowed from<br>Nickel et al. [2016b]  |  |  |
| DNS.md      | TransE   | oui    | its default parameters. RNS results<br>for TransE and RESCAL are<br>borrowed from Nickel et  |  |  |
| DomainNS.md | CTransR  | oui    | 8.7</td> </tr> <tr> <td>CTransR (Lin<br>et al., 2015b)</td> <td>No                           |  |  |
| DomainNS.md | DISTMULT | oui    | 1.4</td> </tr> <tr> <td>DISTMULT<br>(Yang et al., 2015)</td> <td>No                          |  |  |

| Article     | Entité             | Trouvé | Extrait (tableau)                                                                  |  |  |
|-------------|--------------------|--------|------------------------------------------------------------------------------------|--|--|
| DomainNS.md | IRN                | oui    | 4.7</td> </tr> <tr> <td>IRN (Shen et al., 2016)</td> <td>Ex                        |  |  |
| DomainNS.md | ITransF            | oui    | </tr> <tr> <td><strong>ITransF</strong></td> <td>No</td>                           |  |  |
| DomainNS.md | NLFeat             | oui    | 4.6</td> </tr> <tr> <td>NLFeat (Toutanova and Chen, 2015)</td>                     |  |  |
| DomainNS.md | PTransE            | oui    | 6.2</td> </tr> <tr> <td>PTransE (Lin et al., 2015a)</td> <td>Pa                    |  |  |
| DomainNS.md | Random Walk        | oui    | 7.0</td> </tr> <tr> <td>Random Walk (Wei et al., 2016)</td> <td>Pat                |  |  |
| DomainNS.md | RTransE            | oui    | ng></td> </tr> <tr> <td>RTransE (García-Durán et al., 2015)</td>                   |  |  |
| DomainNS.md | STransE            | oui    | the brackets are another set of results STransE reported. <table> <thead> <tr>     |  |  |
| DomainNS.md | TATEC              | oui    | 7.3</td> </tr> <tr> <td>TATEC (García-Durán et al., 2016)</td>                     |  |  |
| DomainNS.md | TransD             | oui    | 4.0</td> </tr> <tr> <td>TransD (Ji et al., 2015)</td> <td>No</                     |  |  |
| DomainNS.md | TransE             | oui    | 6.3</td> </tr> <tr> <td>TransE (Bordes et al., 2013)</td> <td>                     |  |  |
| DomainNS.md | TransH             | oui    | 7.1</td> </tr> <tr> <td>TransH (Wang et al., 2014)</td> <td>No                     |  |  |
| DomainNS.md | TransR             | oui    | 4.4</td> </tr> <tr> <td>TransR (Lin et al., 2015b)</td> <td>No                     |  |  |
| EANS.md     | ComplEx            | oui    | </tr> <tr> <td rowspan="6">ComplEx<br/>(Trouillon et al., 2016)</td>               |  |  |
| EANS.md     | DistMult           | oui    | </tr> <tr> <td rowspan="7">DistMult<br/>(Yang et al., 2014)</td> <t                |  |  |
| EANS.md     | RotatE             | oui    | </tr> <tr> <td rowspan="5">RotatE<br/>(Sun et al., 2019)</td> <td                  |  |  |
| EANS.md     | TransD             | oui    | </tr> <tr> <td rowspan="6">TransD<br/>(Ji et al., 2015)</td> <td>                  |  |  |
| EANS.md     | TransE             | oui    | tbody> <tr> <td rowspan="7">TransE<br/>(Bordes et al., 2013)</td>                  |  |  |
| ERDNS.md    | CompGCN (ConvE)    | oui    | </tr> <tr> <td rowspan="3">CompGCN (ConvE)</td> <td>unif</td> <td                  |  |  |
| ERDNS.md    | CompGCN (DistMult) | oui    | </tr> <tr> <td rowspan="3">CompGCN (DistMult)</td> <td>unif</td> <td               |  |  |
| ERDNS.md    | CompGCN (TransE)   | oui    | </tr> <tr> <td rowspan="3">CompGCN (TransE)</td> <td>unif</td> <td                 |  |  |
| ERDNS.md    | ComplEx            | oui    | 3. Performance comparison on DistMult, ComplEx, TransE and RotatE. <table> <thead> |  |  |
| ERDNS.md    | ConvE              | oui    | le> Table 4. Performance comparison on ConvE and CompGCN. <table> <thead> <tr>     |  |  |

| Article          | Entité      | Trouvé     | Extrait (tableau)                                                                    |  |  |
|------------------|-------------|------------|--------------------------------------------------------------------------------------|--|--|
| ERDNS.md         | DistMult    | oui        | le> Table 3. Performance comparison on DistMult, ComplEx, TransE and RotatE. <table> |  |  |
| ERDNS.md         | RotatE      | oui        | arison on DistMult, ComplEx, TransE and RotatE. <table> <thead> <tr> <th>            |  |  |
| ERDNS.md         | TransE      | oui        | rmance comparison on DistMult, ComplEx, TransE and RotatE. <table> <thead> <tr>      |  |  |
| eTruncatedUNS.md | AlignE      | oui        | 320</td> </tr> <tr> <td>AlignE</td> <td>47.18</td> <td>                              |  |  |
| eTruncatedUNS.md | BootEA      | oui        | 707</td> </tr> <tr> <td>BootEA</td> <td><strong>62.94</strong>                       |  |  |
| eTruncatedUNS.md | IPTransE    | oui        | 334</td> </tr> <tr> <td>IPTransE</td> <td>40.59</td> <td>                            |  |  |
| eTruncatedUNS.md | JAPE        | oui        | 386</td> </tr> <tr> <td>JAPE</td> <td>41.18</td> <td>                                |  |  |
| eTruncatedUNS.md | MTransE     | oui        | </thead> <tbody> <tr> <td>MTransE</td> <td>30.83</td> <td>                           |  |  |
| GHN.md           | DistMult    | oui        | 4.1</td> </tr> <tr> <td>DistMult</td> <td>44.4</td> <td>                             |  |  |
| GHN.md           | DKRL        | oui        | em></td> </tr> <tr> <td>DKRL</td> <td>16.0</td> <td>                                 |  |  |
| GHN.md           | KEPLER      | oui        | es\$</td> </tr> <tr> <td>KEPLER</td> <td>21.0</td> <td>                              |  |  |
| GHN.md           | KG-BERT     | oui        | ods</td> </tr> <tr> <td>KG-BERT</td> <td>21.6</td> <td>                              |  |  |
| GHN.md           | MTL-KGC     | oui        | 2.0</td> </tr> <tr> <td>MTL-KGC</td> <td>33.1</td> <td>                              |  |  |
| GHN.md           | RotatE      | oui        | 4.6</td> </tr> <tr> <td>RotatE</td> <td>47.6</td> <td>                               |  |  |
| GHN.md           | SimKGC      | oui        | 8.2</td> </tr> <tr> <td>SimKGC</td> <td>66.6</td> <td>                               |  |  |
| GHN.md           | StAR        | oui        | 5.8</td> </tr> <tr> <td>StAR</td> <td>40.1</td> <td>                                 |  |  |
| GHN.md           | TransE      | oui        | ods</td> </tr> <tr> <td>TransE</td> <td>24.3</td> <td>                               |  |  |
| GHN.md           | Tucker      | oui        | 3.3</td> </tr> <tr> <td>Tucker</td> <td>47.0</td> <td>                               |  |  |
| GibbsNS.md       | ComplEx     | oui        | 533</td> </tr> <tr> <td>ComplEx</td> <td>0.323</td> <td>                             |  |  |
| GibbsNS.md       | ConvE       | oui        | 526</td> </tr> <tr> <td>ConvE</td> <td>0.325</td> <td>                               |  |  |
| GibbsNS.md       | DistMult    | oui        | d>-</td> </tr> <tr> <td>DistMult</td> <td>0.308</td> <td>                            |  |  |
| GibbsNS.md       | InteractE   | oui        | .52</td> </tr> <tr> <td>InteractE</td> <td>0.354</td> <td>                           |  |  |
| GibbsNS.md       | KGE-EML     | oui        | 569</td> </tr> <tr> <td>KGE-EML [11]</td> <td>0.36</td> <td>                         |  |  |
| GibbsNS.md       | KGE-EML+GNS | <b>NON</b> | -                                                                                    |  |  |
| GibbsNS.md       | PairRE      | oui        | ng></td> </tr> <tr> <td>PairRE</td> <td>0.351</td> <td>                              |  |  |

| Article     | Entité                   | Trouv é    | Extrait (tableau)                                                         |  |  |
|-------------|--------------------------|------------|---------------------------------------------------------------------------|--|--|
| GibbsNS.md  | PROCRUSTES               | oui        | 528</td> </tr> <tr><br><td>PROCRUSTES</td><br><td>0.345</td> <td>         |  |  |
| GibbsNS.md  | RotatE                   | oui        | 529</td> </tr> <tr> <td>RotatE</td><br><td>0.338</td> <td>                |  |  |
| GibbsNS.md  | TransD                   | oui        | </tr> <tr> <td<br>rowspan="5">TransD</td><br><td>Uniform</td> <           |  |  |
| GibbsNS.md  | TransE                   | oui        | </thead> <tbody> <tr><br><td>TransE</td> <td>0.33</td> <td>               |  |  |
| GibbsNS.md  | TuckER                   | oui        | 554</td> </tr> <tr> <td>TuckER</td><br><td>0.358</td> <td>                |  |  |
| GNDN.md     | ConvE                    | oui        | 5.8</td> </tr> <tr> <td>ConvE<br>[19]*</td> <td>0.46</td>                 |  |  |
| GNDN.md     | ConvKB                   | oui        | 9.1</td> </tr> <tr> <td>ConvKB<br>[20]</td> <td>0.248</td>                |  |  |
| GNDN.md     | KBGAN                    | oui        | </thead> <tbody> <tr> <td>KBGAN<br>[19]*</td> <td>0.213</td>              |  |  |
| GNDN.md     | Knowledge Completion GAN | <b>NON</b> | -                                                                         |  |  |
| GNDN.md     | NTN                      | oui        | 5.2</td> </tr> <tr> <td>NTN [8]*</td><br><td>70.6</td>                    |  |  |
| GNDN.md     | Path-KCGAN [2 hop]       | oui        | 4.6</td> </tr> <tr><br><td>Path-KCGAN[2 hop]</td><br><td>0.58</td> <td>   |  |  |
| GNDN.md     | Path-KCGAN [3 hop]       | oui        | 7.1</td> </tr> <tr><br><td>Path-KCGAN[3 hop]</td><br><td>0.60</td> <td>   |  |  |
| GNDN.md     | PathG-RNN [2 hop]        | oui        | </thead> <tbody> <tr><br><td>PathG-RNN [2 hop]</td><br><td>0.41</td> <td> |  |  |
| GNDN.md     | PathG-RNN [3 hop]        | oui        | 6.3</td> </tr> <tr> <td>PathG-RNN [3<br>hop]</td> <td>0.43</td> <td>      |  |  |
| GNDN.md     | PTransE [2 hop]          | oui        | 8.5</td> </tr> <tr> <td>PTransE [2<br>hop]</td> <td>0.49</td> <td>        |  |  |
| GNDN.md     | PTransE [3 hop]          | oui        | 2.2</td> </tr> <tr> <td>PTransE [3<br>hop]</td> <td>0.54</td> <td>        |  |  |
| GNDN.md     | TransE                   | oui        | </thead> <tbody> <tr> <td>TransE<br>[9]*</td> <td>70.9</td>               |  |  |
| GNDN.md     | TransG                   | oui        | 8.9</td> </tr> <tr> <td>TransG<br>[16]*</td> <td>87.4</td>                |  |  |
| GNS.md      | KG-BERT                  | <b>NON</b> | -                                                                         |  |  |
| GraphGAN.md | DeepWalk                 | oui        | </thead> <tbody> <tr><br><td>DeepWalk</td> <td>0.841</td><br><td>         |  |  |
| GraphGAN.md | GraphGAN                 | oui        | 776</td> </tr> <tr><br><td>GraphGAN</td><br><td><strong>0.855</strong>    |  |  |
| GraphGAN.md | LINE                     | oui        | 812</td> </tr> <tr> <td>LINE</td><br><td>0.820</td> <td>                  |  |  |
| GraphGAN.md | Node2vec                 | oui        | 761</td> </tr> <tr><br><td>Node2vec</td> <td>0.845</td><br><td>           |  |  |
| GraphGAN.md | Struc2vec                | oui        | 842</td> </tr> <tr><br><td>Struc2vec</td> <td>0.821</td><br><td>          |  |  |

| Article  | Entité               | Trouv é | Extrait (tableau)                                                                                   |                                                                          |  |  |
|----------|----------------------|---------|-----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|--|--|
| HaSa.md  | ComplEx              | NON     | -                                                                                                   |                                                                          |  |  |
| HaSa.md  | DistMult             | oui     | ng></td> </tr> <tr> <td>DistMult<br>[36]</td> <td>7,000</td>                                        |                                                                          |  |  |
| HaSa.md  | HaSa+                | oui     | /tr> <tr> <td>\$\mathcal{L}_{\text{HaSa}}\$<br>(6)</td> <td><u>123</u></td>                         |                                                                          |  |  |
| HaSa.md  | KBGAN                | oui     | 450</td> </tr> <tr> <td>KBGAN<br>[4]</td> <td>-</td> <td>                                           |                                                                          |  |  |
| HaSa.md  | KG-BERT              | oui     | KGE</td> </tr> <tr> <td>KGBERT<br>[38]</td> <td>97</td> <                                           |                                                                          |  |  |
| HaSa.md  | LASS                 | oui     | 482</td> </tr> <tr> <td>LASS<br>[25]</td> <td>55</td> <                                             |                                                                          |  |  |
| HaSa.md  | RotatE               | oui     | 441</td> </tr> <tr> <td>RotatE<br>[26]</td> <td>3,340</td>                                          |                                                                          |  |  |
| HaSa.md  | sentence-BERT        | oui     | 6\$). All methods use the pre-trained<br>LM sentence-BERT as the<br>initialization. Boldface is the |                                                                          |  |  |
| HaSa.md  | StAR                 | oui     | 420</td> </tr> <tr> <td>StAR<br>[30]</td> <td><strong>51</stron                                     |                                                                          |  |  |
| HaSa.md  | TransE               | oui     | KGE</td> </tr> <tr> <td>TransE<br>[3]</td> <td>2,300</td>                                           |                                                                          |  |  |
| HTENS.md | ComplEx              | oui     | )d\$</td> </tr> <tr> <td>ComplEx</td><br><td>\$&lt; \mathbf{h}, \mat                                |                                                                          |  |  |
| HTENS.md | DistMult             | oui     | )d\$</td> </tr> <tr> <td>DistMult</td><br><td>\$&lt; \mathbf{h}, \mat                               |                                                                          |  |  |
| HTENS.md | TransD               | oui     | )d\$</td> </tr> <tr> <td>TransD</td><br><td>\$\                                                     | (\mathbf{l} + \mat                                                       |  |  |
| HTENS.md | TransE               | oui     | </thead> <tbody> <tr><br><td>TransE</td> <td>\$\                                                    | \mathbf{h} + \math                                                       |  |  |
| IGAN.md  | Single Layer Model   | NON     | -                                                                                                   |                                                                          |  |  |
| IGAN.md  | SME(Bilinear)        | oui     | ng></td> </tr> <tr> <td>SME(Bilinear)<br>(Bordes et al. 2012)</td> <td>                             |                                                                          |  |  |
| IGAN.md  | Structured Embedding | NON     | -                                                                                                   |                                                                          |  |  |
| IGAN.md  | TransD               | oui     | ng></td> </tr> <tr> <td>TransD (Ji et<br>al. 2015)</td> <td>91</t                                   |                                                                          |  |  |
| IGAN.md  | TransE               | oui     | 5.7</td> </tr> <tr> <td>TransE<br>(Bordes et al. 2013)</td> <td><                                   |                                                                          |  |  |
| IGAN.md  | TransE + A-LSTM      | NON     | -                                                                                                   |                                                                          |  |  |
| IGAN.md  | TransH               | oui     | ng></td> </tr> <tr> <td>TransH<br>(Wang et al. 2014)</td> <td>87<                                   |                                                                          |  |  |
| IGAN.md  | TransR               | oui     | ng></td> </tr> <tr> <td>TransR (Lin<br>et al. 2015)</td> <td><str                                   |                                                                          |  |  |
| IGAN.md  | Unstructured Model   | NON     | -                                                                                                   |                                                                          |  |  |
| KBGAN.md | ComplEx              | oui     | re considered in this paper. Except for<br>COMPLEX, all boldface lower case<br>letters repres       |                                                                          |  |  |
| KBGAN.md | DistMult             | oui     | }\                                                                                                  | \$</td> </tr> <tr><br><td>DISTMULT</td><br><td>\$\angle<br>\mathbf{h}, \ |  |  |



| Article           | Entité   | Trouvé | Extrait (tableau)                                                                                 |                                                |                           |
|-------------------|----------|--------|---------------------------------------------------------------------------------------------------|------------------------------------------------|---------------------------|
| KBGAN.md          | TransD   | oui    | } \                                                                                               | \$</td> </tr> <tr><br><td>TRANSD</td> <td>\$\\ | (\mathbf{l} + \mathbf{h}) |
| KBGAN.md          | TransE   | oui    | </thead> <tbody> <tr><br><td>TRANSE</td> <td>\$\\                                                 | \mathbf{h} + \mathbf{h}                        |                           |
| KSGAN.md          | ComplEx  | oui    | ^d\$</td> </tr> <tr> <td>ComplEx</td><br><td>\$Re(\mathbf{h}), \mathbf{h}                         |                                                |                           |
| KSGAN.md          | TransD   | oui    | d)\$</td> </tr> <tr> <td>TransD</td><br><td>\$-\\                                                 | (w_r w_h^{\top} + I)                           |                           |
| KSGAN.md          | TransE   | oui    | br/>distance<br/>model</td><br><td>TransE</td> <td>\$-\\                                          | \mathbf{h} + \mathbf{h}                        |                           |
| LAS.md            | ComplEx  | oui    | 462</td> </tr> <tr> <td>ComplEx<br>(pretrained)</td> <td>0.264</td>                               |                                                |                           |
| LAS.md            | ConvE    | oui    | d>-</td> </tr> <tr> <td>ConvE<br>[23]</td> <td><strong>31.6</str                                  |                                                |                           |
| LAS.md            | DistMult | oui    | </thead> <tbody> <tr> <td>DistMult<br>[18]</td> <td>24.1</td>                                     |                                                |                           |
| LAS.md            | R-GCN    | oui    | 1.0</td> </tr> <tr> <td>R-GCN<br>[28]</td> <td>24.8</td>                                          |                                                |                           |
| LAS.md            | TransD*  | oui    | 431</td> </tr> <tr> <td>TransD<br>(pretrained)</td> <td>0.244</td>                                |                                                |                           |
| LAS.md            | TransE   | oui    | </thead> <tbody> <tr> <td>TransE<br>(pretrained)</td> <td>0.243</td>                              |                                                |                           |
| LEMON.md          | RotatE   | oui    | tbody> <tr> <td<br>rowspan="4">RotatE</td><br><td>Uniform</td> <                                  |                                                |                           |
| LEMON.md          | TransE   | oui    | </tr> <tr> <td<br>rowspan="4">TransE</td><br><td>Uniform</td> <                                   |                                                |                           |
| Localcognitive.md | ComplEx  | oui    | \$\circ\$ denotes element-wise<br>(Hadamard) complex product,<br>\$\otimes\$ denotes element-wise |                                                |                           |
| Localcognitive.md | ConvE    | oui    | 599</td> </tr> <tr> <td>ConvE</td><br><td>374</td> <td>.                                          |                                                |                           |
| Localcognitive.md | DistMult | oui    | ="3">Semantic matching</td><br><td>DistMult</td> <td>\$\angle<br>\mathbf{r}, \mathbf{h}           |                                                |                           |
| Localcognitive.md | HolE     | oui    | d>-</td> </tr> <tr> <td>HolE</td><br><td>-</td> <td>.93                                           |                                                |                           |
| Localcognitive.md | QuatE    | oui    | d>-</td> </tr> <tr> <td>QuatE</td><br><td>\$\mathbf{h} \otimes \mathbf{h}                         |                                                |                           |
| Localcognitive.md | RatE     | oui    | d>\$</td> </tr> <tr> <td>RatE</td><br><td>\$-\\                                                   | \mathbf{h} \odot \mathbf{h}                    |                           |
| Localcognitive.md | RotatE   | oui    | d>\$</td> </tr> <tr> <td>RotatE</td><br><td>\$-\\                                                 | \mathbf{h} \circ \mathbf{h}                    |                           |
| Localcognitive.md | TransE   | oui    | owspan="3">Trans-based</td><br><td>TransE</td> <td>\$-\\                                          | \mathbf{h} + \mathbf{h}                        |                           |
| Localcognitive.md | Tucker   | oui    | Table 9: Performance comparison<br>between Tucker and RatE on<br>WN18RR/FB15k-237. “#0” deno      |                                                |                           |
| LTS.md            | CTransR  | oui    | 8.7</td> </tr> <tr><br><td>CTransR(unif/bern)</td><br><td>243/231</td>                            |                                                |                           |
| LTS.md            | SE       | oui    | ilt</th> </tr> <tr> <td>SE</td><br><td>1011</td> <td>                                             |                                                |                           |

| Article       | Entité      | Trouvé | Extrait (tableau)                                                                   |                                                             |                                        |
|---------------|-------------|--------|-------------------------------------------------------------------------------------|-------------------------------------------------------------|----------------------------------------|
| LTS.md        | TransD      | oui    | 0.2</td> </tr> <tr><br><td>TransD(unif/bern)</td><br><td>242/224</td>               |                                                             |                                        |
| LTS.md        | TransE      | oui    | </thead> <tbody> <tr><br><td>TransE</td> <td>7.01</td> <td>                         |                                                             |                                        |
| LTS.md        | TransE-LTS  | oui    | .71</td> </tr> <tr><br><td>TransE-LTS</td> <td>13.89</td><br><td>                   |                                                             |                                        |
| LTS.md        | TransE-SNS  | oui    | >74</td> </tr> <tr><br><td>TransE-SNS(unif/bern)</td><br><td>220/207</td>           |                                                             |                                        |
| LTS.md        | TransH      | oui    | 7.1</td> </tr> <tr><br><td>TransH(unif/bern)</td><br><td>318/401</td>               |                                                             |                                        |
| LTS.md        | TransSparse | oui    | 7.3</td> </tr> <tr><br><td>TransSparse(unif/bern)</td><br><td>233/223</td>          |                                                             |                                        |
| LTS.md        | TransR      | oui    | 4.4</td> </tr> <tr><br><td>TransR(unif/bern)</td><br><td>232/238</td>               |                                                             |                                        |
| M2ixKG.md     | ComplEx     | oui    | ct, $\bar{\cdot}$ denotes conjugate<br>for complex vectors <table> <thead><br><tr>  |                                                             |                                        |
| M2ixKG.md     | DistMult    | oui    | = "2">semantic matching</td><br><td>DistMult</td> <td> $\angle$<br>$\mathbf{r}$ , \ |                                                             |                                        |
| M2ixKG.md     | RotatE      | oui    | \                                                                                   | $\cdot$ </td> </tr> <tr><br><td>RotatE</td><br><td> $\cdot$ | $\mathbf{h}$<br>} $\cdot$ $\mathbf{m}$ |
| M2ixKG.md     | TransE      | oui    | translational distance</td><br><td>TransE</td> <td> $\cdot$                         | $\mathbf{h}$ + $\mathbf{b}$                                 |                                        |
| MCNS.md       | DeepWalk    | oui    | <tr> <th> </th> <th>DeepWalk</th><br><th>GCN</th> <th>G                             |                                                             |                                        |
| MCNS.md       | GCN         | oui    | > <th>DeepWalk</th> <th>GCN</th><br><th>GraphSAGE</th>                              |                                                             |                                        |
| MCNS.md       | GraphSAGE   | oui    | k</th> <th>GCN</th><br><th>GraphSAGE</th><br><th>DeepWalk</th>                      |                                                             |                                        |
| MDNCaching.md | ComplEx     | oui    | $\cdot$ </td> </tr> <tr> <td>ComplEx<br>[11]</td> <td> $\Re(\cdot)$ diag(           |                                                             |                                        |
| MDNCaching.md | DistMult    | oui    | tic<br>matching-based</td><br><td>DistMult [16]</td> <td> $\cdot$<br>diag(r)        |                                                             |                                        |
| MDNCaching.md | TransD      | oui    | $\cdot$ </td> </tr> <tr> <td>TransD<br>[6]</td> <td> $\cdot$                        | $\mathbf{h} + \mathbf{w}_r \mathbf{w}_\cdot$                |                                        |
| MDNCaching.md | TransE      | oui    | nal<br>distance-based</td><br><td>TransE [2]</td> <td> $\cdot$                      | $\mathbf{h} + \mathbf{r} - \mathbf{t}$                      | $\cdot$                                |
| NMiss.md      | DistMult    | oui    | -06</td> </tr> <tr> <td>DistMult</td><br><td>0.001</td> <td>                        |                                                             |                                        |
| NMiss.md      | RESCAL      | oui    | -06</td> </tr> <tr> <td>Rescal</td><br><td>0.001</td> <td>                          |                                                             |                                        |
| NMiss.md      | TransE      | oui    | 084</td> </tr> <tr> <td>TransE</td><br><td>0.001</td> <td>                          |                                                             |                                        |
| NoiGAN.md     | Attention   | oui    | 507</td> </tr> <tr> <td>Attention</td><br><td><strong>436</strong></td>             |                                                             |                                        |
| NoiGAN.md     | DistMult    | oui    | <td rowspan="10">0%</td><br><td>DistMult</td> <td>.218</td> <td>                    |                                                             |                                        |

| Article      | Entité   | Trouvé | Extrait (tableau)                                                                            |  |  |
|--------------|----------|--------|----------------------------------------------------------------------------------------------|--|--|
| NoiGAN.md    | KBGAN    | oui    | 507</td> </tr> <tr> <td>KBGAN</td><br><td>.266</td> <td>                                     |  |  |
| NoiGAN.md    | RotatE   | oui    | 503</td> </tr> <tr> <td>RotatE</td><br><td>.326</td> <td>                                    |  |  |
| NoiGAN.md    | TransE   | oui    | > </tr> <tr> <td>NoiGAN-TransE<br>(soft)</td> <td><strong>.949</s                            |  |  |
| NSCaching.md | ComplEx  | oui    | op\$</td> </tr> <tr> <td>ComplEx<br>[38]</td> <td>\$\text{Re}(\mathbf{b                      |  |  |
| NSCaching.md | DistMult | oui    | = "2">semantic matching</td><br><td>DistMult [46]</td><br><td>\$\mathbf{h} \cdot             |  |  |
| NSCaching.md | TransD   | oui    | _1\$</td> </tr> <tr> <td>TransD<br>[20]</td> <td>\$\mathbf{h} + \mathbf{v}                   |  |  |
| NSCaching.md | TransE   | oui    | translational distance</td><br><td>TransE [7]</td> <td>\$\mathbf{h} + \mathbf{v}             |  |  |
| NSCaching.md | TransH   | oui    | _1\$</td> </tr> <tr> <td>TransH<br>[42]</td> <td>\$\mathbf{h} - \mathbf{v}                   |  |  |
| PNS.md       | TransE   | oui    | </thead> <tbody> <tr><br><td>TransE</td> <td>263</td> <td>0                                  |  |  |
| PNS.md       | TransH   | oui    | 349</td> </tr> <tr><br><td>TransH(bern)</td> <td>401</td>                                    |  |  |
| PNS.md       | TransR   | oui    | <th>Epoch</th> <th>Modified<br>TransR</th> <th>Original<br>TransR</th>                       |  |  |
| RAAKGC.md    | ComplEx  | oui    | 9.2</td> </tr> <tr> <td>ComplEx</td><br><td>46.8</td> <td>                                   |  |  |
| RAAKGC.md    | ConvE    | oui    | 5.9</td> </tr> <tr> <td>ConvE</td><br><td>45.6</td> <td>                                     |  |  |
| RAAKGC.md    | DistMult | oui    | 9.7</td> </tr> <tr> <td>DistMult</td><br><td>44.4</td> <td>                                  |  |  |
| RAAKGC.md    | HaSa     | oui    | 4.9</td> </tr> <tr> <td>HaSa</td><br><td>53.8</td> <td>                                      |  |  |
| RAAKGC.md    | InsKGC   | oui    | s for the text-based methods, we<br>choose InsKGC (Jiang, Drummond,<br>and Cohn 2023) and Si |  |  |
| RAAKGC.md    | KG-BERT  | oui    | ods</td> </tr> <tr><br><td>KG-BERT</td> <td>21.6</td><br><td>                                |  |  |
| RAAKGC.md    | MTL-KGC  | oui    | d>.</td> </tr> <tr><br><td>MTL-KGC</td> <td>33.1</td><br><td>                                |  |  |
| RAAKGC.md    | QuatE    | oui    | 9.0</td> </tr> <tr> <td>QuatE</td><br><td>48.1</td> <td>                                     |  |  |
| RAAKGC.md    | RAA-KGC  | oui    | d>.</td> </tr> <tr><br><td>RAA-KGC</td><br><td><strong><u>59.74</u></td>                     |  |  |
| RAAKGC.md    | RGCN     | oui    | d>.</td> </tr> <tr> <td>RGCN</td><br><td>42.7</td> <td>                                      |  |  |
| RAAKGC.md    | RotatE   | oui    | d>.</td> </tr> <tr> <td>RotatE</td><br><td>47.6</td> <td>                                    |  |  |
| RAAKGC.md    | SimKGC   | oui    | d>.</td> </tr> <tr> <td>SimKGC</td><br><td>55.31</td> <td>                                   |  |  |
| RAAKGC.md    | StAR     | oui    | d>.</td> </tr> <tr> <td>StAR</td><br><td>40.1</td> <td>                                      |  |  |
| RAAKGC.md    | TransE   | oui    | ods</td> </tr> <tr> <td>TransE</td><br><td>24.3</td> <td>                                    |  |  |

| Article          | Entité    | Trouvé | Extrait (tableau)                                                                            |  |  |
|------------------|-----------|--------|----------------------------------------------------------------------------------------------|--|--|
| RandomCorrupt.md | ComplEx   | oui    | modifications to adapt this task; (2) <b>ComplEx</b> achieves the best performance on all    |  |  |
| RandomCorrupt.md | DistMult  | oui    | ificantly and consistently outperforms <b>DistMult</b> which matches the observations for th |  |  |
| RandomCorrupt.md | TransE    | oui    | of link prediction; (2) Surprisingly, <b>TransE</b> has competitive performance with **Co    |  |  |
| RCWC.md          | ComplEx   | oui    | 946</td> </tr> <tr> <td>ComplEx [Trouillon et al., 2016]</td> <                              |  |  |
| RCWC.md          | ConvE     | oui    | 947</td> </tr> <tr> <td>ConvE [Dettmers et al., 2018]</td> <t                                |  |  |
| RCWC.md          | DistMult  | oui    | 943</td> </tr> <tr> <td>DistMult [Yang et al., 2014]</td> <td>42                             |  |  |
| RCWC.md          | InteractE | oui    | .54</td> </tr> <tr> <td>InteractE [Vashishth et al., 2020]</td> <                            |  |  |
| RCWC.md          | KGBoost-R | oui    | 951</td> </tr> <tr> <td>KGBoost-R (Ours)</td> <td>16</td>                                    |  |  |
| RCWC.md          | KGBoost-T | oui    | ng></td> </tr> <tr> <td>KGBoost-T (Ours)</td> <td><strong>15</str                            |  |  |
| RCWC.md          | RotatE    | oui    | 956</td> </tr> <tr> <td>RotatE [Sun et al., 2019]</td> <td>40<                               |  |  |
| RCWC.md          | SACN      | oui    | ng></td> </tr> <tr> <td>SACN [Shang et al., 2019]</td> <td>-                                 |  |  |
| RCWC.md          | TransE    | oui    | </thead> <tbody> <tr> <td>TransE [Bordes et al., 2013]</td> <td>                             |  |  |
| ReasonKGE.md     | ComplEx   | oui    | </tr> <tr> <td rowspan="3">ComplEx</td> <td>LUBM3U</td> <t                                   |  |  |
| ReasonKGE.md     | TransE    | oui    | tbody> <tr> <td rowspan="3">TransE</td> <td>LUBM3U</td> <t                                   |  |  |
| RUGA.md          | ComplEx   | oui    | 546</td> </tr> <tr> <td>ComplEx</td> <td>0.690</td> <td>                                     |  |  |
| RUGA.md          | DistMult  | oui    | 468</td> </tr> <tr> <td>DistMult</td> <td>0.634</td> <td>                                    |  |  |
| RUGA.md          | HolE      | oui    | 570</td> </tr> <tr> <td>HolE</td> <td>0.608</td> <td>                                        |  |  |
| RUGA.md          | RUGA      | oui    | </tr> <tr> <td><strong>RUGA</strong></td> <td><strong>0.779                                  |  |  |
| RUGA.md          | RUGE      | oui    | 601</td> </tr> <tr> <td>RUGE</td> <td>0.767</td> <td>                                        |  |  |
| RUGA.md          | TransE    | oui    | </thead> <tbody> <tr> <td>TransE</td> <td>h, t</td> <td>                                     |  |  |
| SANS.md          | DistMult  | oui    | </tr> <tr> <td rowspan="6">DistMult</td> <td>KBGAN</td> <td>                                 |  |  |
| SANS.md          | RotatE    | oui    | </tr> <tr> <td rowspan="4">RotatE</td> <td>Uniform</td> <                                    |  |  |
| SANS.md          | TransE    | oui    | tbody> <tr> <td rowspan="6">TransE</td> <td>KBGAN (Cai and Wang, 2                           |  |  |

| Article      | Entité             | Trouvé | Extrait (tableau)                                                                        |                              |                                    |
|--------------|--------------------|--------|------------------------------------------------------------------------------------------|------------------------------|------------------------------------|
| SelfAdv.md   | ComplEx            | oui    | $\overline{\cdot}$ denotes conjugate for complex vectors, and 2D reshaping for real vect |                              |                                    |
| SelfAdv.md   | ConvE              | oui    | s, and 2D reshaping for real vectors in ConvE model. TransX represents a wide range o    |                              |                                    |
| SelfAdv.md   | DistMult           | oui    | $R^k$                                                                                    | DistMult (Yang et al., 2014) |                                    |
| SelfAdv.md   | HolE               | oui    | $C^k$                                                                                    | HolE (Nickel et al., 2016)   |                                    |
| SelfAdv.md   | pRotatE            | oui    | .956                                                                                     | pRotatE                      | 43                                 |
| SelfAdv.md   | RotatE             | oui    | $R^k$                                                                                    | RotatE                       |                                    |
| SelfAdv.md   | TransE             | oui    | odel. TransX represents a wide range of TransE's variants, such as TransH (Wang et al.   |                              |                                    |
| SNS.md       | DistMult           | oui    | $^2$                                                                                     | DistMult [12]                |                                    |
| SNS.md       | TransH             | oui    |                                                                                          | TransH [11]                  |                                    |
| SparseNSG.md | ANALOGY            | oui    | 0.1                                                                                      | ANALOGY                      | $d = 200, \alpha = 0.1$            |
| SparseNSG.md | ComplEx            | oui    | 0.1                                                                                      | COMPLEX                      | $d = 200, \alpha = 0.1$            |
| SparseNSG.md | DistMult           | oui    | 001                                                                                      | DISTMULT                     | $d = 200, \alpha = 0.1$            |
| SparseNSG.md | TransE             | oui    |                                                                                          | TransE                       | $d=100, \gamma = 1, \alpha = 0.00$ |
| SparseNSG.md | TransH             | oui    | 001                                                                                      | TransH                       | $d = 100, \gamma = 1, \alpha = 0.$ |
| STC.md       | LFM                | oui    | 1.3                                                                                      | LFM                          | 283                                |
| STC.md       | RESCAL             | oui    |                                                                                          | RESCAL                       | 828                                |
| STC.md       | SE                 | oui    | 4.1                                                                                      | SE                           | 273                                |
| STC.md       | SME (bilinear)     | oui    | 0.8                                                                                      | SME (bilinear)               | 284                                |
| STC.md       | SME (linear)       | oui    | 9.8                                                                                      | SME (linear)                 | 274                                |
| STC.md       | TKRL (RHE)         | oui    | 7.2                                                                                      | TKRL (RHE)                   | 184                                |
| STC.md       | TKRL (RHE+STC)     | NON    | -                                                                                        |                              |                                    |
| STC.md       | TKRL (RHE+STC+TCE) | NON    | -                                                                                        |                              |                                    |
| STC.md       | TKRL (WHE)         | oui    | 9.4                                                                                      | TKRL (WHE)                   | 186                                |

| Article            | Entité                      | Trouvé | Extrait (tableau)                                                         |               |           |
|--------------------|-----------------------------|--------|---------------------------------------------------------------------------|---------------|-----------|
| STC.md             | TKRL (WHE+STC)              | NON    | -                                                                         |               |           |
| STC.md             | TKRL (WHE+STC+TCE)          | NON    | -                                                                         |               |           |
| STC.md             | TransE                      | oui    | 3.1</td> </tr> <tr> <td>TransE</td><br><td>238</td> <td>1                 |               |           |
| STC.md             | TransE+STC+TCE              | NON    | -                                                                         |               |           |
| STC.md             | TransE+TCE                  | NON    | -                                                                         |               |           |
| STC.md             | TransR                      | oui    | 2.1</td> </tr> <tr> <td>TransR</td><br><td>199</td> <td>7                 |               |           |
| STC.md             | TransR+STC+TCE              | NON    | -                                                                         |               |           |
| STC.md             | TransR+TCE                  | NON    | -                                                                         |               |           |
| TANS.md            | Complex                     | oui    | <td>FB15k-237</td><br><td>Complex</td> <td>25.0</td><br><td>              |               |           |
| TANS.md            | DistMult                    | oui    | <td>FB15k-237</td><br><td>DistMult</td> <td>24.0</td> <td>                |               |           |
| TANS.md            | HAKE                        | oui    | <td>FB15k-237</td> <td>HAKE</td><br><td>33.0</td> <td>                    |               |           |
| TANS.md            | HousE                       | oui    | <td>FB15k-237</td> <td>HousE</td><br><td>33.5</td> <td>                   |               |           |
| TANS.md            | RotatE                      | oui    | <td>FB15k-237</td> <td>RotatE</td><br><td>33.0</td> <td>                  |               |           |
| TANS.md            | TransE                      | oui    | <td>FB15k-237</td><br><td>TransE</td> <td>33.0</td> <td>                  |               |           |
| TruncatedNS.md     | IPTransE                    | oui    | 258</td> </tr> <tr><br><td>IPTransE</td> <td>32.73</td><br><td>           |               |           |
| TruncatedNS.md     | JAPE                        | oui    | 412</td> </tr> <tr> <td>JAPE</td><br><td>14.45</td> <td>                  |               |           |
| TruncatedNS.md     | MTransE                     | oui    | 007</td> </tr> <tr> <td>MtransE</td><br><td>16.51</td> <td>               |               |           |
| TruncatedNS.md     | TransE                      | oui    | </thead> <tbody> <tr><br><td>TransE</td> <td>8.35</td> <td>               |               |           |
| TruncatedNS.md     | Truncated Negative Sampling | NON    | -                                                                         |               |           |
| TuckerDNCaching.md | Complex                     | oui    | ^n\$</td> </tr> <tr> <td>Complex</td><br><td>\$Re(h^top \cdot diag(       |               |           |
| TuckerDNCaching.md | DistMult                    | oui    | tic<br>matching-based</td><br><td>DistMult</td> <td>\$h^top \cdot diag(r) |               |           |
| TuckerDNCaching.md | TransD                      | oui    | ^n\$</td> </tr> <tr> <td>TransD</td><br><td>\$\left\                      | h + w_r w_h^t |           |
| TuckerDNCaching.md | TransE                      | oui    | nal<br>distance-based</td><br><td>TransE</td> <td>\$\                     | h + r - t\    | _{1/2}\$< |
| TypeAugmented.md   | Complex                     | oui    | td> </tr> <tr><br><td>Take+Complex</td> <td>O(d + k)</td>                 |               |           |
| TypeAugmented.md   | DistMult                    | oui    | td> </tr> <tr><br><td>Take+Distmult</td> <td>O(d + k)</td>                |               |           |

| Article            | Entité                | Trouvé | Extrait (tableau)                                                                      |  |  |
|--------------------|-----------------------|--------|----------------------------------------------------------------------------------------|--|--|
| TypeAugmented.md   | RotatE                | oui    | $\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u}^T \mathbf{W} \mathbf{v}$          |  |  |
| TypeAugmented.md   | Simple                | oui    | $\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u}^T \mathbf{W} \mathbf{v}$          |  |  |
| TypeAugmented.md   | Take-Complex          | oui    | $\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u}^T \mathbf{W} \mathbf{v}$          |  |  |
| TypeAugmented.md   | Take-DistMult         | oui    | $\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u}^T \mathbf{W} \mathbf{v}$          |  |  |
| TypeAugmented.md   | Take-RotatE           | oui    | $\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u}^T \mathbf{W} \mathbf{v}$          |  |  |
| TypeAugmented.md   | Take-Simple           | oui    | $\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u}^T \mathbf{W} \mathbf{v}$          |  |  |
| TypeAugmented.md   | Take-TransE           | oui    | $\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u}^T \mathbf{W} \mathbf{v}$          |  |  |
| TypeAugmented.md   | TransE                | oui    | $\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u}^T \mathbf{W} \mathbf{v}$          |  |  |
| TypeAugmented.md   | TypeComplex           | oui    | $\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u}^T \mathbf{W} \mathbf{v}$          |  |  |
| TypeAugmented.md   | TypeDM                | oui    | $\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u}^T \mathbf{W} \mathbf{v}$          |  |  |
| TypeConstraints.md | mwNN                  | oui    | omparison of AUPRC and AUROC result for mwNN [8] with and without exploiting prior k   |  |  |
| TypeConstraints.md | RESCAL                | oui    | omparison of AUPRC and AUROC result for RESCAL with and without exploiting prior knowl |  |  |
| TypeConstraints.md | TransE                | oui    | omparison of AUPRC and AUROC result for TransE with and without exploiting prior knowl |  |  |
| Uniform.md         | LFM                   | oui    | $\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u}^T \mathbf{W} \mathbf{v}$          |  |  |
| Uniform.md         | RESCAL                | oui    | $\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u}^T \mathbf{W} \mathbf{v}$          |  |  |
| Uniform.md         | SME(bilinear)         | oui    | $\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u}^T \mathbf{W} \mathbf{v}$          |  |  |
| Uniform.md         | SME(linear)           | oui    | $\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u}^T \mathbf{W} \mathbf{v}$          |  |  |
| Uniform.md         | Structured Embeddings | NON    | -                                                                                      |  |  |
| Uniform.md         | TransE                | oui    | $\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u}^T \mathbf{W} \mathbf{v}$          |  |  |
| Uniform.md         | Unstructured          | oui    | $\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u}^T \mathbf{W} \mathbf{v}$          |  |  |

## B. Preuve - MENTIONNÉS seulement (238)

| Article         | Entité   | Trouvé | Extrait (prose)                                                                                     |
|-----------------|----------|--------|-----------------------------------------------------------------------------------------------------|
| AdaptativeNS.md | HAKE     | oui    | relationships in knowledge graphs. The HAKE [20] model is designed for graph embedd                 |
| AdaptativeNS.md | MEI      | oui    | ent intricacies of the knowledge graph. MEI [13] and MEIM [14] partition each embed                 |
| AdaptativeNS.md | MEIM     | oui    | es of the knowledge graph. MEI [13] and MEIM [14] partition each embedding into mult                |
| AdaptativeNS.md | RESCAL   | oui    | ressed in vector space representations. RESCAL [10] introduced a tensor factorization               |
| AdaptativeNS.md | TransD   | oui    | between entities and relationships. The TransD [18] model further improves upon TransR              |
| AdaptativeNS.md | TransG   | oui    | and relationships more effectively. The TransG [17] model proposes a solution to the c              |
| AdaptativeNS.md | TransH   | oui    | many and many-to-one relationships, the TransH [16] model extends it by introducing re              |
| AdaptativeNS.md | TransR   | oui    | omes the shortcomings of TransE. In the TransR [7] model, a relation mapping matrix is              |
| ADNS.md         | ComplEx  | oui    | t years. e.g, TransE [2], DistMult [3], Complex[4], TransH [5], RotatE [6] etc. Almost              |
| ADNS.md         | RotatE   | oui    | , DistMult [3], Complex[4], TransH [5], RotatE [6] etc. Almost all of these models dep              |
| ADNS.md         | TransH   | oui    | , TransE [2], DistMult [3], Complex[4], TransH [5], RotatE [6] etc. Almost all of thes              |
| BatchNS.md      | DistMult | oui    | n operators allows PBG to train RESCAL, DistMult, TransE, and ComplEx models (Nickel et             |
| BatchNS.md      | word2vec | oui    | this literature are algorithms such as word2vec which allowed word embedding methods to             |
| CAKE.md         | ComplEx  | oui    | se together with the characteristics of complex relations. Furthermore, a multi-view li             |
| CAKE.md         | DistMult | oui    | osition-based** score function, such as DistMult (Yang et al., 2015): $\$E(h, r, t) = \backslash m$ |
| CAKE.md         | JOIE     | oui    | hand, although some KGE models such as JOIE (Hao et al., 2019) employ the ontology                  |



| Article | Entité    | Trouvé | Extrait (prose)                                                                          |
|---------|-----------|--------|------------------------------------------------------------------------------------------|
| CAKE.md | KBGAN     | oui    | 2014). (2) Adversarial-based sampling: KBGAN (Cai and Wang, 2018) integrates the KGE     |
| CAKE.md | RESCAL    | oui    | s such as TransE (Bordes et al., 2013), RESCAL (Nickel et al., 2011), ComplEx (Trouill   |
| CAKE.md | TransH    | oui    | amely 1-1, 1-N, N-1, and N-N defined in TransH (Wang et al., 2014) for negative sampli   |
| CANS.md | GTrans    | oui    | iple as a manifold rather than a point. GTrans [20] used dynamic and static weighting    |
| CANS.md | ManifoldE | oui    | ctors of entities into various spaces. ManifoldE [19] embedded a triple as a manifold ra |
| CANS.md | puTransE  | oui    | d the noise from other relation spaces. puTransE [21] generated multiple embedding space |
| CANS.md | TransD    | oui    | ame vector space. Then TransR [17], and TransD [18], extended TransE, by projecting th   |
| CANS.md | TransH    | oui    | th 1-to-N, N-to-1, or N-to-N relations. TransH [10] proposed to project the entity to    |
| CANS.md | TransR    | oui    | ons were in the same vector space. Then TransR [17], and TransD [18], extended TransE,   |
| CCS.md  | HAKE      | oui    | on translation of the vectors in space. HAKE [31] maps the entities into spatial pol     |
| CCS.md  | NTN       | oui    | SME) [16] model, neural tensor network (NTN) [17] model, and matrix factorization (      |
| CCS.md  | OTE       | oui    | ntities into spatial polar coordinates. OTE [32] exploits the orthogonality relatio      |
| CCS.md  | PTransE   | oui    | TransE [26], TransA [27], TransAt [28], PTransE [29], etc. Recently, RotatE [30] has ch  |
| CCS.md  | RESCAL    | oui    | ) [17] model, and matrix factorization (RESCAL) [18] model. However, these KGE models    |
| CCS.md  | RotatE    | oui    | nsAt [28], PTransE [29], etc. Recently, RotatE [30] has changed the translation proces   |
| CCS.md  | SE        | oui    | tion, such as the structured embedding (SE) [15] model, semantic matching energy (       |
| CCS.md  | SME       | oui    | ) [15] model, semantic matching energy (SME) [16] model, neural tensor network (NTN      |

| Article            | Entité     | Trouvé | Extrait (prose)                                                                           |
|--------------------|------------|--------|-------------------------------------------------------------------------------------------|
| CCS.md             | STransE    | oui    | nsD [23], TransM [24], TranSparse [25], STransE [26], TransA [27], TransAt [28], PTrans   |
| CCS.md             | TransA     | oui    | sM [24], TranSparse [25], STransE [26], TransA [27], TransAt [28], PTransE [29], etc.     |
| CCS.md             | TransAt    | oui    | Sparse [25], STransE [26], TransA [27], TransAt [28], PTransE [29], etc. Recently, Rota   |
| CCS.md             | TransM     | oui    | TransH [21], TransR [22], TransD [23], TransM [24], TranSparse [25], STransE [26], Tr     |
| CCS.md             | TranSparse | oui    | TransR [22], TransD [23], TransM [24], TranSparse [25], STransE [26], TransA [27], TransA |
| CondConstraints.md | RESCAL     | oui    | nly explores their applicability to the RESCAL model [2]. On the other hand, the work     |
| CondConstraints.md | TransH     | oui    | cheme was introduced by Wang et al. for TransH and is sometimes called the Bernoulli t    |
| CondConstraints.md | TRESCAL    | oui    | exploiting type information [6]. While TRESCAL explicitly tries to make use of type co    |
| dans.md            | RGCN       | oui    | relational graph convolutional network (RGCN) [26] to serve as our base embedding mo      |
| dans.md            | TransE     | oui    | xt{China} \rangle\$ and a classic method TransE [2]. TransE maps each entity and relati   |
| DeMix.md           | TransH     | oui    | amely 1-N, N-1, 1-1, and N-N defined in TransH [24], we record the num of positive tri    |
| DHNS.md            | KGDM       | oui    | idering the query node's context, while KGDM [16] applies DDPM to estimate the proba      |
| DHNS.md            | LAFA       | oui    | y in an adversarial training framework. LAFA [24] considers the relationships between     |
| DHNS.md            | QuatE      | oui    | ntisymmetric relations. RotatE [25] and QuatE [36] introduce rotational and quaternion    |
| DHNS.md            | RESCAL     | oui    | ar interaction-based KGE models such as RESCAL [22] and DistMult [35], formulated as:     |
| DHNS.md            | TransH     | oui    | based KGE models such as TransE [3] and TransH [31], formulated as: $\mathbf{x}_e$ ,      |

| Article          | Entité                | Trouvé | Extrait (prose)                                                                                                 |
|------------------|-----------------------|--------|-----------------------------------------------------------------------------------------------------------------|
| DHNS.md          | VISTA                 | oui    | aggregation of multi-modal information. VISTA [11] designs three transformer-based en                           |
| DNS.md           | ComplEx               | oui    | ued embeddings into their corresponding complex variants, and performs Hermitian dot pr                         |
| DNS.md           | HolE                  | oui    | teractions within the triples for their HolE system. Trouillon et al. [2016] represe                            |
| DNS.md           | Neural Tensor Network | oui    | h as <i>RESCAL</i> Nickel et al. [2011] and <i>Neural Tensor Network</i> Socher et al. [2013] employ higher ord |
| DNS.md           | TransH                | oui    | Wang et al. [2014] define an algorithm <i>TransH</i> that behaves like <i>TransE</i> with a mino                |
| EANS.md          | RESCAL                | oui    | relation vectors into various spaces. RESCAL (Nickel et al., 2011), DistMult (Yang e                            |
| EANS.md          | TransH                | oui    | . Various extensions of TransE, such as TransH (Wang et al., 2014), TransR (Lin et al.                          |
| EANS.md          | TransR                | oui    | sE, such as TransH (Wang et al., 2014), TransR (Lin et al., 2015) and TransD (Ji et al                          |
| ERDNS.md         | CompGCN               | oui    | ], ConvE [5], and InteractE [24], while CompGCN [5] is a generalized version of several                         |
| ERDNS.md         | ConvKB                | oui    | odels. Examples of CNN-based models are ConvKB [18], ConvE [5], and InteractE [24], wh                          |
| ERDNS.md         | InteractE             | oui    | models are ConvKB [18], ConvE [5], and InteractE [24], while CompGCN [5] is a generalize                        |
| ERDNS.md         | SimpleE               | oui    | ples and lower scores to negative ones. Simple negative sampling methods like uniform                           |
| ERDNS.md         | TransD                | oui    | amples of TD models include TransE [2], TransD [8], TransR [15], and RotatE [21], whil                          |
| ERDNS.md         | TransR                | oui    | models include TransE [2], TransD [8], TransR [15], and RotatE [21], while popular SM                           |
| eTruncatedUNS.md | KR-EAR                | oui    | tes reverse triples and relation paths. KR-EAR [Lin et al., 2016] introduces categoric                          |
| eTruncatedUNS.md | PTransE               | oui    | 2014], TransR [Lin et al., 2015b] and PTransE [Lin et al., 2015a]. Most existing KG                             |
| eTruncatedUNS.md | TransE                | oui    | nherent KG semantics. As an early work, TransE [Bordes et al., 2013] interprets a rela                          |

| Article          | Entité    | Trouvé | Extrait (prose)                                                                          |
|------------------|-----------|--------|------------------------------------------------------------------------------------------|
| eTruncatedUNS.md | TransH    | oui    | further improved by many works, such as TransH [Wang et al., 2014], TransR [Lin et al.   |
| eTruncatedUNS.md | TransR    | oui    | s, such as TransH [Wang et al., 2014], TransR [Lin et al., 2015b] and PTransE [Lin et    |
| GHN.md           | Complex   | oui    | al., 2013), TransH (Wang et al., 2014), Complex (Trouillon et al., 2016), and RotatE (S  |
| GHN.md           | TransH    | oui    | s include TransE (Bordes et al., 2013), TransH (Wang et al., 2014), Complex (Trouillon   |
| GibbsNS.md       | KG-BERT   | oui    | ed pre-trained models to tackle issues. KG-BERT [14], for instance, is a BERT-based kno  |
| GibbsNS.md       | RESCAL    | oui    | . An eminent contender in this class is RESCAL proposed by Nickel et al. [24], which c   |
| GibbsNS.md       | TransH    | oui    | gh similarity. To overcome challenges, TransH, introduced by [9], utilizes hyperplane    |
| GNDN.md          | DistMult  | oui    | ions over entities and relations, and a DistMult based decoder for factorization. The GC |
| GNDN.md          | ELECTRA   | oui    | ecently developed language model called ELECTRA [37], which is a compute-efficient GAN-  |
| GNDN.md          | HypER     | oui    | igm and reports better overall results. HypER [26] uses a 1D-relation-specific convol    |
| GNDN.md          | KCGAN     | oui    | iscriminative belief prediction models. KCGAN thus invokes a game between generator-n    |
| GNDN.md          | KG-BERT   | oui    | ansformer-based encoding models such as KG-BERT [35] are inspired by transformer-based   |
| GNDN.md          | PTransE   | oui    | red as PathG-RNNs. We also compare with PTransE which is a variant of TransE for integr  |
| GNS.md           | BERT      | oui    | state-of-the-art embedding method - KG-BERT for triple classification task - on a b      |
| GNS.md           | Complex   | oui    | n KGs such as: TransE [5], Rescal [12], Complex [16], DistMult [20] and KG-BERT [21], a  |
| GNS.md           | DistMult  | oui    | TransE [5], Rescal [12], Complex [16], DistMult [20] and KG-BERT [21], and all these me  |
| GNS.md           | ReasonKGE | oui    | o pick head or tail entity to corrupt. ReasonKGE [9] with its Iterative Negative Samplin |

| Article     | Entité                | Trouvé | Extrait (prose)                                                                                          |
|-------------|-----------------------|--------|----------------------------------------------------------------------------------------------------------|
| GNS.md      | Rescal                | oui    | embeddings in KGs such as: TransE [5], Rescal [12], ComplEx [16], DistMult [20] and K                    |
| GNS.md      | TransE                | oui    | and relation embeddings in KGs such as: TransE [5], Rescal [12], ComplEx [16], DistMul                   |
| GNS.md      | TransOWL              | oui    | e sampling for triple classification 5 TransOWL [7] leverages ontology axioms in order                   |
| GraphGAN.md | PPNE                  | oui    | nder the supervision of edge existence. PPNE (Li et al. 2017b) directly learns verte                     |
| GraphGAN.md | SDNE                  | oui    | aining data in the graph. For instance, SDNE (Wang, \*M. Guo is the corresponding a                      |
| IGAN.md     | Neural Tensor Network | oui    | ag{8}\$\$ where g() is tanh function. <b>Neural Tensor Network</b> (Socher et al. 2013) extends Single L |
| KBGAN.md    | ConvE                 | oui    | o combine the two entities in a triple. CONVE (Dettmers et al., 2017) uses a convolut                    |
| KBGAN.md    | HoIE                  | oui    | iple as a manifold rather than a point. HOLE (Nickel et al., 2016) employs circular                      |
| KBGAN.md    | ManifoldE             | oui    | ent models achieve strong performances. MANIFOLDE (Xiao et al., 2016) embeds a triple as                 |
| KBGAN.md    | RESCAL                | oui    | graph embedding (KGE) techniques (e.g., RESCAL (Nickel et al., 2011), TRANSE (Bordes e                   |
| KBGAN.md    | TransH                | oui    | ased embedding. Later variants, such as TRANSH (Wang et al., 2014), TRANSR (Lin et al.                   |
| KBGAN.md    | TransR                | oui    | ts, such as TRANSH (Wang et al., 2014), TRANSR (Lin et al., 2015) and TRANSD (Ji et al                   |
| KSGAN.md    | DistMult              | oui    | ng model (e.g., TransE [6], TransD [7], DistMult [8], ComplEx [9]). However, previous w                  |
| KSGAN.md    | GTrans                | oui    | (or $\mathbf{t} - \mathbf{r}$ ). As a generic model, GTrans [17] introduces eigenstate and mimesis       |
| KSGAN.md    | ManifoldE             | oui    | ict magnitude constraints between them. ManifoldE [16] uses a manifold function to constr                |
| KSGAN.md    | MLP                   | oui    | ic matching models such as NTN [21] and MLP [22], focus on neural network architect                      |
| KSGAN.md    | NTN                   | oui    | Other semantic matching models such as NTN [21] and MLP [22], focus on neural netw                       |

| Article  | Entité   | Trouvé | Extrait (prose)                                                                           |
|----------|----------|--------|-------------------------------------------------------------------------------------------|
| KSGAN.md | RESCAL   | oui    | from the translational distance model, RESCAL [18] is a classic semantic matching mod     |
| KSGAN.md | SimpleE  | oui    | complex space rather than a real space. SimpleE [19] simplifies ComplEx by considering    |
| KSGAN.md | TransF   | oui    | ons. Targeting at more flexible models, TransF [15] ensures that $t$ (or $h$ ) has        |
| KSGAN.md | TransH   | oui    | $t$ $(h, r, t)$ . Various variants such as TransH [12], TransM [13], TransR [14] and Tran |
| KSGAN.md | TransM   | oui    | . Various variants such as TransH [12], TransM [13], TransR [14] and TransD [7] have b    |
| KSGAN.md | TransR   | oui    | iants such as TransH [12], TransM [13], TransR [14] and TransD [7] have been developed    |
| KSGAN.md | Tucker   | oui    | different similarity scoring function. Tucker [20] is based on Tucker decomposition a     |
| LAS.md   | ANALOGY  | oui    | bability-based models, e.g. ProjE [21], ANALOGY [22] and R-GCN [28], also achieve good    |
| LAS.md   | DKRL     | oui    | ] exploit relation paths, TEKE [33] and DKRL [34] embed knowledge graph with textual      |
| LAS.md   | HoIE     | oui    | dels (e.g., RESCAL [17], DistMult [18], HoIE [19], ComplEx [20], and ConvE [23] etc)      |
| LAS.md   | ITransF  | oui    | [13], STransE [14], puTransE [15], and ITransF [16]) define an energy function accordi    |
| LAS.md   | KALE     | oui    | nowledge graph with textual information, KALE [35] and RUGE [36] utilize logical rule     |
| LAS.md   | KG2E     | oui    | 6], TransH [7], TransR [8], TransD [9], KG2E [10], TransA [11], TransSparse [12], Tra     |
| LAS.md   | ProjE    | oui    | ny other probability-based models, e.g. ProjE [21], ANALOGY [22] and R-GCN [28], also     |
| LAS.md   | PTransE  | oui    | and TKRL [30] consider the entity type, PTransE [31] and RTransE [32] exploit relation    |
| LAS.md   | puTransE | oui    | Sparse [12], TransG [13], STransE [14], puTransE [15], and ITransF [16]) define an energ  |
| LAS.md   | RESCAL   | oui    | models Probability-based models (e.g., RESCAL [17], DistMult [18], HoIE [19], ComplEx     |

| Article           | Entité                                  | Trouvé     | Extrait (prose)                                                                                                       |
|-------------------|-----------------------------------------|------------|-----------------------------------------------------------------------------------------------------------------------|
| LAS.md            | RTransE                                 | oui        | sider the entity type, PTransE [31] and RTransE [32] exploit relation paths, TEKE [33]                                |
| LAS.md            | RUGE                                    | oui        | with textual information, KALE [35] and RUGE [36] utilize logical rules. In this pa                                   |
| LAS.md            | SSE                                     | oui        | further improve the task. For example, SSE [29] and TKRL [30] consider the entity                                     |
| LAS.md            | STransE                                 | oui        | nsA [11], TransSparse [12], TransG [13], STransE [14], puTransE [15], and ITransF [16])                               |
| LAS.md            | TEKE                                    | oui        | nd RTransE [32] exploit relation paths, TEKE [33] and DKRL [34] embed knowledge grap                                  |
| LAS.md            | TKRL                                    | oui        | ove the task. For example, SSE [29] and TKRL [30] consider the entity type, PTransE                                   |
| LAS.md            | TransA                                  | oui        | [7], TransR [8], TransD [9], KG2E [10], TransA [11], TransSparse [12], TransG [13], STr                               |
| LAS.md            | TransG                                  | oui        | G2E [10], TransA [11], TransSparse [12], TransG [13], STransE [14], puTransE [15], and                                |
| LAS.md            | TransH                                  | oui        | slation-based models (e.g., TransE [6], TransH [7], TransR [8], TransD [9], KG2E [10],                                |
| LAS.md            | TranSparse                              | oui        | 8], TransD [9], KG2E [10], TransA [11], TransSparse [12], TransG [13], STransE [14], puTran                           |
| LAS.md            | TransR                                  | oui        | d models (e.g., TransE [6], TransH [7], TransR [8], TransD [9], KG2E [10], TransA [11]                                |
| Localcognitive.md | Relation-adaptive translating Embedding | oui        | 10.04863v1 [cs.CL] 10 Oct 2020 # RatE: Relation-Adaptive Translating Embedding for Knowledge Graph Completion **Hao H |
| Localcognitive.md | TorusE                                  | <b>NON</b> | -                                                                                                                     |
| Localcognitive.md | TransD                                  | <b>NON</b> | -                                                                                                                     |
| Localcognitive.md | TransH                                  | <b>NON</b> | -                                                                                                                     |
| Localcognitive.md | TransR                                  | <b>NON</b> | -                                                                                                                     |
| LTS.md            | GAN                                     | oui        | c distribution sampling, represented by GAN[2], is used to provide high quality neg                                   |
| MCNS.md           | FastGCN                                 | oui        | lleviate the receptive field expansion. FastGCN [5] adopts importance sampling in each                                |

| Article      | Entité                 | Trouvé | Extrait (prose)                                                                                          |
|--------------|------------------------|--------|----------------------------------------------------------------------------------------------------------|
| MCNS.md      | LINE                   | oui    | Embedding methods (e.g. DeepWalk [27], LINE [32]) and Graph Neural Networks (e.g. G                      |
| MCNS.md      | node2vec               | oui    | s, such as DeepWalk [27], LINE [32] and node2vec [11], actually assign each node two emb                 |
| NMiss.md     | ComplEx                | oui    | methods - RESCAL, TransE, DistMult and ComplEx - and evaluate on benchmark datasets -                    |
| NMiss.md     | HoIE                   | oui    | s as well as the Holographic Embedding (HoIE) model, so HoIE was not included<sup>1<                     |
| NMiss.md     | Neural Tensor Networks | oui    | ns. Methods such as <b>Rescal</b> [21] and Neural Tensor Networks [24] learn millions of parameters that |
| NoiGAN.md    | CKRL                   | oui    | 19)), (3) noise aware KGE models (e.g., CKRL (Xie et al., 2018)) and (4) KGE models                      |
| NoiGAN.md    | ComplEx                | oui    | 2011), DistMult (Yang et al., 2014) and ComplEx (Trouillon et al., 2016). To optimize t                  |
| NoiGAN.md    | RESCAL                 | oui    | esentations. The typical models include RESCAL (Nickel et al., 2011), DistMult (Yang e                   |
| NoiGAN.md    | TransH                 | oui    | relation. TransE (Bordes et al., 2013), TransH (Wang et al., 2014) and TransR (Lin et                    |
| NoiGAN.md    | TransR                 | oui    | , 2013), TransH (Wang et al., 2014) and TransR (Lin et al., 2015) are the representati                   |
| NSCaching.md | ANALOGY                | oui    | DistMult [46], HoIE [31], ComplEx [38], ANALOGY [27], etc. All these methods are summar                  |
| NSCaching.md | HoIE                   | oui    | e variants of RESCAL are DistMult [46], HoIE [31] and ComplEx [38]. DistMult simplif                     |
| NSCaching.md | ManifoldE              | oui    | nsD [20], TransSparse [21], TransM [11], ManifoldE [45], etc., and semantic matching model               |
| NSCaching.md | RESCAL                 | oui    | e plausibility of triplets $(h, r, t)$ . RESCAL [32] is the most original model. The en                  |
| NSCaching.md | TransM                 | oui    | nsR [26], TransD [20], TransSparse [21], TransM [11], ManifoldE [45], etc., and semanti                  |
| NSCaching.md | TranSparse             | oui    | TransH [42], TransR [26], TransD [20], TransSparse [21], TransM [11], ManifoldE [45], etc.               |
| NSCaching.md | TransR                 | oui    | his problem, variants like TransH [42], TransR [26], TransD [20] are introduced to pro                   |



| Article          | Entité       | Trouvé | Extrait (prose)                                                                                 |
|------------------|--------------|--------|-------------------------------------------------------------------------------------------------|
| PNS.md           | TransG       | oui    | o the current state-of-the-art approach TransG[4] to further improve its performance.           |
| RAAKGC.md        | BERTRL       | oui    | lation-aware neighborhood of the query. BERTRL (Zha, Chen, and Yan 2022) collects poss          |
| RAAKGC.md        | ConKGC       | oui    | he query as extra context to the model. ConKGC (Shang et al. 2023) introduces a sampli          |
| RAAKGC.md        | DKRL         | oui    | . 2024). The pioneer text-based method, DKRL (Xie et al. 2016), initially integrates            |
| RAAKGC.md        | GenKGC       | oui    | sion. Besides the mentioned algorithms, GenKGC (Xie et al. 2022) and KICGPT (Wei et al          |
| RAAKGC.md        | KEPLER       | oui    | ng KG-BERT (Yao, Mao, and Luo 2019) and KEPLER (Wang et al. 2021b), utilize a Pre-trai          |
| RAAKGC.md        | KG-S2S       | oui    | KGT5-context (Kochsiek et al. 2023) and KG-S2S (Chen et al. 2022a) model the link pred          |
| RAAKGC.md        | KGT5-context | oui    | r encoding text descriptions. Moreover, KGT5-context (Kochsiek et al. 2023) and KG-S2S (Chen    |
| RAAKGC.md        | KICGPT       | oui    | lgorithms, GenKGC (Xie et al. 2022) and KICGPT (Wei et al. 2023) go beyond the neighbo          |
| RAAKGC.md        | LMKE         | oui    | s by the decoder directly. Furthermore, LMKE (Wang et al. 2022b) introduces a contra            |
| RandomCorrupt.md | ConvE        | oui    | combine the two entities in a triple. <b>ConvE</b> (Dettmers et al., 2017) uses a convol        |
| RandomCorrupt.md | HOLE         | oui    | <b>DistMult</b> into the complex space. <b>HOLE</b> (Nickel et al., 2016) employs circula       |
| RandomCorrupt.md | ManifoldE    | oui    | tors of entities into various spaces. <b>ManifoldE</b> (Xiao et al., 2016) embeds a triple a    |
| RandomCorrupt.md | PTransE      | oui    | ), <b>TransR</b> (Lin et al., 2015b) and <b>PTransE</b> (Lin et al., 2015a). However, limited   |
| RandomCorrupt.md | RESCAL       | oui    | le as a manifold rather than a point. <b>RESCAL</b> (Nickel et al., 2011) is one of the e       |
| RandomCorrupt.md | TransD       | oui    | ), <b>TransR</b> (Lin et al., 2015b) and <b>TransD</b> (Ji et al., 2015), extend <b>TransE</b>  |
| RandomCorrupt.md | TransH       | oui    | ed embedding. Later variants, such as <b>TransH</b> (Wang et al., 2014), <b>TransR</b> (Lin     |
| RandomCorrupt.md | TransR       | oui    | ch as <b>TransH</b> (Wang et al., 2014), <b>TransR</b> (Lin et al., 2015b) and <b>TransD</b> (J |

| Article      | Entité       | Trouvé | Extrait (prose)                                                                                           |
|--------------|--------------|--------|-----------------------------------------------------------------------------------------------------------|
| RCWC.md      | RESCAL       | oui    | trix $M_r$ is used to model a relation. RESCAL [Nickel et al., 2011] suffers from mode                    |
| RCWC.md      | TransH       | oui    | $r \approx 0$ for symmetric relations. TransH [Lin et al., 2015] and TransR [Lin et a                     |
| RCWC.md      | TransR       | oui    | elations. TransH [Lin et al., 2015] and TransR [Lin et al., 2015] model symmetric rela                    |
| ReasonKGE.md | Embed2Reason | oui    | mpling. For example, a related method <i>Embed2Reason</i> ( <i>E2R</i> ) has been proposed by Garg *et al |
| RUGA.md      | RESCAL       | oui    | coring function. Typical models include RESCAL <sup>[13]</sup> , DistMult <sup>[14]</sup>                 |
| RUGA.md      | TransD       | oui    | H <sup>[9]</sup> , TransR <sup>[10]</sup> , TransD <sup>[11]</sup> and so on. As shown in F               |
| RUGA.md      | TransH       | oui    | ne to expand and apply TranE, including TransH <sup>[9]</sup> , TransR <sup>[10]</sup> , T                |
| RUGA.md      | TransR       | oui    | TranE, including TransH <sup>[9]</sup> , TransR <sup>[10]</sup> , TransD <sup>[11]</sup> a                |
| SelfAdv.md   | KG2E         | oui    | We also compare an additional approach KG2E_KL (He et al., 2015), which is a probab                       |
| SelfAdv.md   | STransE      | oui    | 2014), TransR (Lin et al., 2015b), and STansE (Nguyen et al., 2016), where $g_{r,i}(\cdot)$               |
| SelfAdv.md   | TorusE       | oui    | and concurrent work to our work is the TorusE (Ebisu & Ichise, 2018) model, which def                     |
| SelfAdv.md   | TransH       | oui    | ide range of TransE's variants, such as TransH (Wang et al., 2014), TransR (Lin et al.                    |
| SelfAdv.md   | TransR       | oui    | ts, such as TransH (Wang et al., 2014), TransR (Lin et al., 2015b), and STansE (Nguye                     |
| SparseNSG.md | NTN          | oui    | other kinds of models, such as SE, SME, NTN. However, a practical knowledge graph i                       |
| SparseNSG.md | SE           | oui    | 15], and other kinds of models, such as SE, SME, NTN. However, a practical knowled                        |
| SparseNSG.md | SME          | oui    | ter contains, Semantic Matching Energy (SME) [10], Semantically Smooth Embedding (S                       |
| SparseNSG.md | SSE          | oui    | E) [10], Semantically Smooth Embedding (SSE) [11], and translation-based methods [1                       |

| Article      | Entité                                          | Trouvé | Extrait (prose)                                                                                                            |
|--------------|-------------------------------------------------|--------|----------------------------------------------------------------------------------------------------------------------------|
| SparseNSG.md | TransA                                          | oui    | sing, some presentation models, such as TransA [16], TransG [19], replace the relation                                     |
| SparseNSG.md | TransD                                          | oui    | ls, such as TransE, TransH, TransR, and TransD [15], and other kinds of models, such a                                     |
| SparseNSG.md | TransG                                          | oui    | esentation models, such as TransA [16], TransG [19], replace the relation of a triplet                                     |
| SparseNSG.md | TranSparse                                      | oui    | ment, widely used in TransR, TransD and TranSparse [18], can effectively reduce the probab                                 |
| SparseNSG.md | TransR                                          | oui    | s, such as TransE [12], TransH [13] and TransR [14], are the most used knowledge graph                                     |
| STC.md       | DKRL                                            | oui    | s, entity names or entity descriptions. DKRL [Xie <i>et al.</i> , 2016] proposes descripti                                 |
| STC.md       | NTN                                             | oui    | triples, is significant for RL in KGs. NTN [Socher <i>et al.</i> , 2013] encodes 3-way t                                   |
| STC.md       | SME                                             | oui    | 2011; 2012], SE [Bordes <i>et al.</i> , 2011], SME [Bordes <i>et al.</i> , 2012; 2014] and LFM [Je                         |
| STC.md       | TransD                                          | oui    | judging the distance between entities. TransD [Ji <i>et al.</i> , 2015] proposes dynamic ma                                |
| STC.md       | TransH                                          | oui    | o-1 and N-to-N. To address this issue, TransH [Wang <i>et al.</i> , 2014b] interprets relat                                |
| STC.md       | Type-embodied Knowledge Representation Learning | oui    | paper, we propose a novel method named Type-embodied Knowledge Representation Learning (TKRL) to take advantages of hie... |
| TANS.md      | BERT                                            | oui    | s, Clark <i>et al.</i> (2020a) indicate that a BERT (Devlin <i>et al.</i> , 2019)-like model ELECTR                        |
| TANS.md      | ELECTRA                                         | oui    | a BERT (Devlin <i>et al.</i> , 2019)-like model ELECTRA (Clark <i>et al.</i> , 2020b) uses the NS loss                     |
| TANS.md      | GenKGC                                          | oui    | ods like KGT5 (Saxena <i>et al.</i> , 2022) and GenKGC (Xie <i>et al.</i> , 2022) are unique in direct                     |
| TANS.md      | KEPLER                                          | oui    | guage model (PLM)-based approaches like KEPLER (Wang <i>et al.</i> , 2021) and SimKGC (Wang <i>et</i>                      |
| TANS.md      | KGT5                                            | oui    | ning. Generation-based KGC methods like KGT5 (Saxena <i>et al.</i> , 2022) and GenKGC (Xie <i>e</i>                        |
| TANS.md      | SimKGC                                          | oui    | hes like KEPLER (Wang <i>et al.</i> , 2021) and SimKGC (Wang <i>et al.</i> , 2022) also have an import                     |

| Article            | Entité    | Trouvé     | Extrait (prose)                                                                          |
|--------------------|-----------|------------|------------------------------------------------------------------------------------------|
| TANS.md            | word2vec  | oui        | n et al. (2019) import subsampling from word2vec (Mikolov et al., 2013) to KGE. Subsampl |
| TruncatedNS.md     | BootEA    | oui        | ame, relationship, and attribute views. BootEA [24] proposes iterative entity alignmen   |
| TruncatedNS.md     | CrossE    | oui        | sed in the entity triple, respectively. CrossE [13] believes that bidirectional effect   |
| TruncatedNS.md     | DAT       | oui        | y updated using newly aligned entities. DAT [21] encodes both entity names and enti      |
| TruncatedNS.md     | HAKE      | oui        | modelled entities in hyperbolic space. HAKE [16] believes that polar coordinates ar      |
| TruncatedNS.md     | KD-CoE    | oui        | been abstracted to related data types. KD-CoE [26] jointly learns multi-lingual entit    |
| TruncatedNS.md     | MultiKE   | oui        | function of the interaction embeddings. MultiKE [23] uses multiple views of entities to  |
| TruncatedNS.md     | MuRP      | oui        | d as rotation changes in complex space. MuRP [15] points out that the relationships      |
| TruncatedNS.md     | RESCAL    | oui        | ions to gauge the rationality of facts. RESCAL [11] was the earliest semantic matching   |
| TruncatedNS.md     | RotatE    | oui        | ntities and relationships. According to RotatE [14], which can model and infer differe   |
| TruncatedNS.md     | SimpleE   | oui        | ons between latent factors of entities. SimpleE [12] restricts the relationship matrix   |
| TruncatedNS.md     | TransD    | oui        | wo different projection matrices by the TransD [17] model, respectively. According to    |
| TruncatedNS.md     | TransEdge | oui        | erforms well in handling tail entities. TransEdge [22] interacts with embeddings between |
| TruncatedNS.md     | TransH    | oui        | n the corresponding relationship space. TransH [19] thinks an entity can have many rep   |
| TruncatedNS.md     | TransR    | oui        | [17] model, respectively. According to TransR [18], each entity is made up of several    |
| TuckerDNCaching.md | RESCAL    | <b>NON</b> | -                                                                                        |
| TuckerDNCaching.md | RotatE    | oui        | ich has an impact on efficiency. Later, RotatE (Sun et al., 2019) introduced a self-ad   |

| Article            | Entité                | Trouvé | Extrait (prose)                                                                                                                       |
|--------------------|-----------------------|--------|---------------------------------------------------------------------------------------------------------------------------------------|
| TuckerDNCaching.md | TorusE                | oui    | lational distance-based models, such as TorusE (Ebisu et al., 2018), by addressing the                                                |
| TuckerDNCaching.md | TuckER                | oui    | ng: high-quality negative sampling with tucker decomposition Tiroshan Madushanka <sup>&lt;sup&gt;</sup>                               |
| TypeAugmented.md   | CAKE                  | oui    | esides, Niu et al. <sup>&lt;sup&gt;33&lt;/sup&gt; propose CAKE which automatically extract commonsense</sup>                          |
| TypeAugmented.md   | HAKE                  | oui    | g to tail embedding. Soon after RotatE, HAKE <sup>&lt;sup&gt;16&lt;/sup&gt; and RatE<sup>&lt;sup&gt;17&lt;/sup&gt; make</sup></sup>   |
| TypeAugmented.md   | JOIE                  | oui    | ons to model the semantic transitivity. JOIE <sup>&lt;sup&gt;24&lt;/sup&gt; directly represents a KG a</sup>                          |
| TypeAugmented.md   | RatE                  | oui    | oon after RotatE, HAKE <sup>&lt;sup&gt;16&lt;/sup&gt; and RatE<sup>&lt;sup&gt;17&lt;/sup&gt; make slight improvements t</sup></sup>   |
| TypeAugmented.md   | TaKE                  | oui    | ed Knowledge graph Embedding framework (TaKE) which could utilize type features to e                                                  |
| TypeAugmented.md   | TaRP                  | oui    | two views. Both TransT <sup>&lt;sup&gt;30&lt;/sup&gt; and TaRP<sup>&lt;sup&gt;32&lt;/sup&gt; collect relation types fro</sup></sup>   |
| TypeAugmented.md   | TKRL                  | oui    | xploring the usage of type information. TKRL <sup>&lt;sup&gt;22&lt;/sup&gt; extends traditional TransE</sup>                          |
| TypeAugmented.md   | TransC                | oui    | nct representations in different types. TransC <sup>&lt;sup&gt;23&lt;/sup&gt; clearly distinguish concep</sup>                        |
| TypeAugmented.md   | TransH                | oui    | of TransE have been developed. Such as, TransH <sup>&lt;sup&gt;13&lt;/sup&gt; projects entity embeddings</sup>                        |
| TypeAugmented.md   | TransR                | oui    | stead of relation-specific hyperplanes, TransR <sup>&lt;sup&gt;14&lt;/sup&gt; directly projects entity e</sup>                        |
| TypeAugmented.md   | TransT                | oui    | d jointly encodes these two views. Both TransT <sup>&lt;sup&gt;30&lt;/sup&gt; and TaRP<sup>&lt;sup&gt;32&lt;/sup&gt; coll</sup></sup> |
| TypeConstraints.md | Neural Tensor Network | oui    | deling of KGs. [20] recently proposed a neural tensor network, which we did not consider in our study                                 |
| TypeConstraints.md | TransH                | oui    | network proposed in [20]. [23] proposed TransH which improves TransE's capability to m                                                |

| Article            | Entité              | Trouvé | Extrait (prose)                                                                                    |
|--------------------|---------------------|--------|----------------------------------------------------------------------------------------------------|
| TypeConstraints.md | TransR              | oui    | further extended in [14] by introducing TransR which separates representations of enti             |
| Uniform.md         | Neural Tensor Model | oui    | point. Another related approach is the Neural Tensor Model [14]. A special case of this model corr |

## C1. Candidats faux négatifs - ÉVALUÉS (38)

| Article     | Entité       | Où          | Extrait                                                                                 |
|-------------|--------------|-------------|-----------------------------------------------------------------------------------------|
| ASA.md      | GCN          | prose+table | egating the neighbors' information. <b>R-GCN</b> (Schlichtkrull et al. 2018) considers  |
| BatchNS.md  | GCN          | prose+table | al., 2018). The problem studied by the GCN is different than the one solved by PBG      |
| CAKE.md     | CAKE         | prose+table | Xiv:2202.13785v3 [cs.AI] 17 Apr 2022 # CAKE: A Scalable Commonsense-Aware Framework     |
| dans.md     | CAKE         | prose+table | nerate informative negative triplets; <b>CAKE</b> [19]: a framework which leverages ext |
| dans.md     | KBGAN        | prose+table | ous state-of-the-art approaches such as KBGAN [3], we adopt a generative adversarial    |
| DHNS.md     | KBGAN        | prose+table | nerate higher-quality negative triples. KBGAN [4] and IGAN [30] use Generative Advers   |
| DHNS.md     | OTE          | prose+table | x [27], RotatE [25], PairRE [7], and GC-OTE [26]. (2) <b>Multimodal KGC models:</b> w   |
| DomainNS.md | KG2E         | table       | 0.2</td> </tr> <tr> <td>KG2E (He et al., 2015)</td> <td>No</td>                         |
| DomainNS.md | NTN          | table       | 6.7</td> </tr> <tr> <td>NTN (Socher et al., 2013)</td> <td>                             |
| DomainNS.md | SE           | table       | </thead> <tbody> <tr> <td>SE (Bordes et al., 2011)</td> <td>                            |
| DomainNS.md | Unstructured | table       | 9.8</td> </tr> <tr> <td>Unstructured (Bordes et al., 2014)</td> <td>                    |
| EANS.md     | KBGAN        | table       | 501</td> </tr> <tr> <td>KBGAN(Cai and Wang, 2017)<sup>††</sup></td>                     |
| ERDNS.md    | KBGAN        | prose+table | natural language processing (NLP) [7]. KBGAN [3], IGAN [26], and NSCaching [34] crea    |
| GibbsNS.md  | KBGAN        | prose+table | is of static negative sampling methods, KBGAN [17] found that the majority of negativ   |

| Article       | Entité       | Où          | Extrait                                                                                       |
|---------------|--------------|-------------|-----------------------------------------------------------------------------------------------|
| HTENS.md      | KBGAN        | prose+table | The GAN-based sampling methods, such as KBGAN[3] and IGAN[13], set the KGE model to b         |
| IGAN.md       | SE           | prose+table | results show that 1. Models including SE, SME(bilinear), TransE and TransH are f              |
| IGAN.md       | SME          | prose+table | ults show that 1. Models including SE, SME(bilinear), TransE and TransH are furthe            |
| IGAN.md       | Unstructured | prose+table | ages (Xie et al. 2016a) of entities. <b>Unstructured Model</b> (Bordes et al. 2012; Bordes et |
| KBGAN.md      | KBGAN        | prose+table | Xiv:1711.04071v3 [cs.CL] 16 Apr 2018 # KBGAN: Adversarial Learning for Knowledge Gra          |
| KSGAN.md      | KBGAN        | prose+table | ed by an adversarial learning framework KBGAN, this paper proposes a new knowledge se         |
| LAS.md        | GCN          | prose+table | ls, e.g. ProjE [21], ANALOGY [22] and R-GCN [28], also achieve good results for lin           |
| LAS.md        | KBGAN        | prose+table | h outperforms GAN-based sampling method KBGAN. <b>INDEX TERMS</b> Knowledge graph, know       |
| LEMON.md      | KBGAN        | table       | 1)\$</td> </tr> <tr> <td>KBGAN [6]</td> <td>\$\mathcal{O}(t)\$</                              |
| LTS.md        | LFM          | table       | 1.3</td> </tr> <tr> <td>LFM</td> <td>469</td> <td>4                                           |
| LTS.md        | SME          | table       | 9.8</td> </tr> <tr> <td>SME</td> <td>542/526</td> <                                           |
| M2ixKG.md     | KBGAN        | prose+table | tribution. IGAN [Wang et al., 2018] and KBGAN [Cai and Wang, 2018] introduce generati         |
| MCNS.md       | KBGAN        | prose+table | ptively generate hard negative samples. KBGAN [3] employs a sampling-based adaptation         |
| MDNCaching.md | KBGAN        | prose+table | ative sampling strategies IGAN [12] and KBGAN [3] were introduced to generate negativ         |
| NSCaching.md  | KBGAN        | prose+table | wo pioneered works, i.e., IGAN [39] and KBGAN [9], attempting to address these challe         |
| ReasonKGE.md  | ReasonKGE    | prose+table | we present a novel iterative approach <i>ReasonKGE</i> that identifies dynamically via symbol |
| SANS.md       | KBGAN        | prose+table | egative sampling (Bordes et al., 2013), KBGAN (Cai and Wang, 2018), and NSCaching (Zh         |

| Article            | Entité | Où          | Extrait                                                                               |
|--------------------|--------|-------------|---------------------------------------------------------------------------------------|
| SelfAdv.md         | KBGAN  | prose+table | en from (Cai (Cai & Wang, 2017), where KBGAN uses a ComplEx negative sample generato  |
| SelfAdv.md         | SE     | table       | </thead> <tbody> <tr> <td>SE (Bordes et al., 2011)</td> <td>\$-                       |
| SNS.md             | KBGAN  | prose+table | trained to learn embeddings. IGAN [8], KBGAN [3], KSGAN [7] are three existing GAN-b  |
| STC.md             | TKRL   | prose+table | died Knowledge Representation Learning (TKRL) to take advantages of hierarchical ent  |
| TuckerDNCaching.md | KBGAN  | prose+table | ributions, IGAN (Wang et al., 2018) and KBGAN (Cai et al., 2018) were introduced as n |
| Uniform.md         | SE     | prose+table | those of [3] (Structured Embeddings or SE) and [14]. 3 Table 1: **Numbers of pa       |
| Uniform.md         | SME    | prose+table | 2], and the energy-based models SE [3], SME(linear)/SME(bilinear) [2] and LFM [6].    |

## C2. Candidats faux négatifs - MENTIONNÉS (11)

| Article            | Entité    | Où    | Extrait                                                                                  |
|--------------------|-----------|-------|------------------------------------------------------------------------------------------|
| ConceptDriven.md   | CAKE      | prose | Commonsense-Aware Knowledge Embedding (CAKE) framework, which consists of automatic      |
| DeMix.md           | CAKE      | prose | CANS [16]. The CANS is a component of CAKE [16] responsible for solving the invali       |
| DeMix.md           | KBGAN     | prose | triples with high scores. For example, KBGAN[6] and IGAN[23] introduce a generative      |
| DMNS.md            | GraphSAGE | prose | conduct experiments using GAT [45] and GraphSAGE (SAGE) [14]. For GCN, we employ two lay |
| GNDN.md            | GCN       | prose | sists of a Convolutional Graph Network (GCN) [39] based encoder for learning repres      |
| HTENS.md           | TKRL      | prose | <sup>1</sup><sup>https://github.com/thunlp/TKRL SIGIR '23, July 23-27, 2023, Taipei, T   |
| Localcognitive.md  | KBGAN     | prose | effectively learn structured knowledge. KBGAN (Cai and Wang, 2018) uses knowledge gra    |
| NSCaching.md       | LINE      | prose | EGATIVE ENTITIES IN CACHE ON FB13. EACH LINE DISPLAYS 5 RANDOM SAMPLED NEGATIVE ENTI     |
| RAAKGC.md          | KGT5      | prose | r encoding text descriptions. Moreover, KGT5-context (Kochsiek et al. 2023) and KG-S     |
| RUGA.md            | KBGAN     | prose | herefore, Liwei Cai et al.[16] proposed KBGAN to solve the problem that negative samp    |
| TypeConstraints.md | SME       | prose | sup>9</sup> https://github.com/glorotxa/SME <sup>10</sup> Mainly caused by the rank      |